

**Q1** Which of the following is not an object Oriented Programming language ?

- (A) Pearl (B) Python  
(C) Small talk (D) SQL Plus

**Q2** Shift-Reduce conflicts are resolved by :

- (A) Left recursion  
(B) Left factoring  
(C) Ambiguity  
(D) Associativity and Precedence

**Q3** A least integer  $n$  such that  $f(x)$  is  $O(x^n)$  for each of the following functions. Arrange the following according to the value of  $n$  in increasing order:

- A.  $f(x)=3x^3+(\log x)^4$   
B.  $f(x)=(x^5+x^2+1)/(x^3+1)$   
C.  $f(x)=(x^3+x^2+1)/(x^2+1)$   
D.  $f(x)=(x^2+5\log x)/(x^2+1)$

Choose the correct answer from the options given below :

- (A) D, C, A, B  
(B) D, C, B, A  
(C) B, C, A, D  
(D) B, A, C, D

**Q4** Arrange the following so that they can be placed in perfect/appropriate order.

- A. Planning  
C. Modeling  
B. Construction  
E. Deployment  
D. Communication

Choose the correct answer from the options given below :

- (A) A, B, C, D, E  
(B) D, A, C, B, E  
(C) A, D, B, C, E  
(D) B, C, D, A, E

**Q5** Match List-I with List-II :

| List - I                                          | List-II                     |
|---------------------------------------------------|-----------------------------|
| A. $A \rightarrow aB \mid a, a \in T, A, B \in V$ | I. Recursive Descent Parser |

|                                                      |                           |
|------------------------------------------------------|---------------------------|
| B. $A \rightarrow BC \mid a, a \in T, A, B, C \in V$ | II. Turing Machine        |
| C. LL (1) grammar                                    | III. Choamsky Normal Form |
| D. Halting problem                                   | IV. Finite Automate       |

Choose the correct answer from the options given below :

- (A) A-IV, B-III, C-I, D-II  
(B) A-III, B-I, C-II, D-IV  
(C) A-II, B-IV, C-III, D-I  
(D) A-IV, B-III, C-II, D-I

**Q6** What is a perceptron in ANN ?

- (A) A double layer auto-associative neural network.  
(B) A neural network that contains feedback.  
(C) An auto-associative neural network.  
(D) A single layer feed-forward neural network with pre-processing.

**Q7** Arrange the following examples of Artificial Intelligence (AI) in the order of increasing complexity :

- A. Spam email detection using rule-based systems.  
B. Handwritten digit recognition using shallow neural networks.  
C. Image classification using convolutional neural networks.  
D. Autonomous driving using reinforcement learning algorithms.

Choose the correct answer from the options given below :

- (A) A, B, C, D  
(B) B, A, C, D  
(C) A, B, D, C  
(D) B, C, A, D

**Q8** Match List-I with List-II :

| List-I Line coding scheme | List-II Bandwidth (average) |
|---------------------------|-----------------------------|
| A. NRZ-L                  | I. $B = N/4$                |
| B. 2B1Q                   | II. $B = N/8$               |



|              |              |
|--------------|--------------|
| C. 4.D-PAM 5 | III. $B=N/3$ |
| D. MLT-3     | IV. $B= N/2$ |

Choose the correct answer from the options given below :

- (A) A-II, B-I, C-IV, D-III  
 (B) A-IV, B-II, C-I, D-III  
 (C) A-IV, B-I, C-II, D-III  
 (D) A-III, B-IV, C-II, D-I

**Q9** The order of steps of Address Translation in Linux Virtual Memory Scheme will be :

- A. register  
 B. offset  
 C. Page directory.  
 D. Page table  
 E. Page Middle directory

Choose the correct answer from the options given below :

- (A) B, A, D, E, C (B) A, C, E, D, B  
 (C) D, C, A, B, E (D) C, A, D, E, B

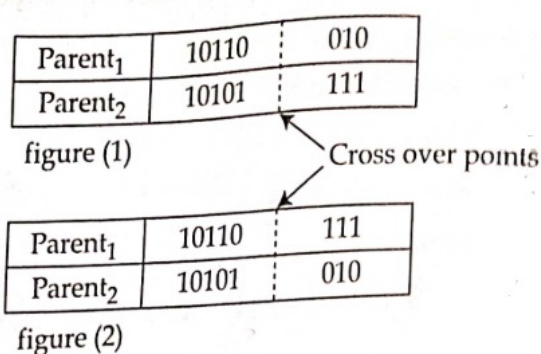
**Q10** If  $f(n) = 5f(n/2) + 3$  and  $f(1) = 7$  is a recurrence relation where K is positive integer, then  $f(2^K) =$

- (A)  $5^K (31/4) - 3/4$  (B)  $5^K (29/4)$   
 (C)  $2^K (31/4) - 3/4$  (D)  $2^K (29/4)$

**Q11** The instruction LD ADR (X) which is equivalent to  $AC \leftarrow M[ADR+XR]$  is in :

- (A) Direct Addressing Mode  
 (B) Indirect Addressing Mode  
 (C) Index Addressing Mode  
 (D) Immediate Addressing Mode

**Q12** For the following Parents in figure (1) and (2) :



Which is correct?

- (A)

figure (2) is the two point crossover of figure (1)

- (B) figure (2) is one point crossover of figure (1)  
 (C) figure (2) is N point crossover of figure (1)  
 (D) figure (2) is uniform crossover of figure (1)

**Q13** The sequence of steps for a page replacement algorithm will be :

- A. find a frame that is not currently being used  
 B. change the page table  
 C. use a freed frame  
 D. no frame is free  
 E. free a frame

Choose the correct answer from the options given below :

- (A) D, A, E, B, C (B) E, B, A, C, D  
 (C) B, E, A, D, C (D) D, E, B, A, C

**Q14** Match List - I with List - II.

$f(n) = a f(n/b) + g(n)$  is a divide-and-conquer recurrence relation for the Listed-I algorithms, match their recurrence in Listed-II.

| List - I Divide-and-conquer based algorithms | List - II Recurrence Relation |
|----------------------------------------------|-------------------------------|
| A. Binary Search                             | I. $f(n) = 7f(n/2) + 15n^2/4$ |
| B. Find maximum and minimum of a sequence    | II. $f(n) = 2f(n/2) + n$      |
| C. Merge Sort                                | III. $f(n) = 2f(n/2) + 2$     |
| D. Fast Matrix Multiplication                | IV. $f(n) = f(n/2) + 2$       |

Choose the correct answer from the options given below :

- (A) A-I, B-II, C-III, D-IV  
 (B) A-II, B-III, C-IV, D-I  
 (C) A-III, B-IV, C-I, D-II  
 (D) A-IV, B-III, C-II, D-I

**Q15** Which of the following control problem does not exist, when processes unaware of each other during interaction?

- (A) Mutual exclusion



- (B) Data coherence  
(C) Starvation  
(D) Deadlock

**Q16** The height of a binary search tree for the words banana, peach, apple, pear, coconut, mango and papaya using alphabetical order is :

- (A) 2 (B) 3  
(C) 4 (D) 5

**Q17** Match List-I with List-II :

| List - I                  | List - II                                             |
|---------------------------|-------------------------------------------------------|
| A. Batch Multiprogramming | I. allows more efficient use of main memory           |
| B. Time sharing           | II. user no longer has direct access to the processor |
| C. Monitor                | III. Maximize processor use                           |
| D. Reentrant Procedures   | IV. minimize response time                            |

Choose the correct answer from the options given below :

- (A) A-III, B-II, C-IV, D-I  
(B) A-III, B-IV, C-II, D-I  
(C) A-I, B-III, C-IV, D-II  
(D) A-II, B-I, C-IV, D-III

**Q18** Which of the following is the function of the semantic analysis phase of the compilation process?

- (A) Type conversion  
(B) Tokenization  
(C) Loop optimization  
(D) Data flow Analysis

**Q19** For the regular languages and context-free languages which is not correct?

- A. both are closed under union operation  
B. both are closed under concatenation operation  
C. both are closed under intersection operation  
D. both are closed under complementation operation  
E. both are closed under Kleen star operation

Choose the most appropriate answer from the options given below :

- (A) A and B Only (B) B and C Only  
(C) C and D Only (D) D and E only

**Q20** Match List-I with List-II :

| List-I                                      | List-II                    |
|---------------------------------------------|----------------------------|
| A. Back Tracking                            | I. Automata                |
| B. Infinite languages with matching numbers | II. Undecidable Problem    |
| C. Canonical LR Parser                      | III. Predictive Parser     |
| D. Post Correspondence Problem              | IV. Large number of states |

Choose the correct answer from the options given below:

- (A) A-I, B-II, C-III, D-IV  
(B) A-II, B-III, C-I, D-III  
(C) A-IV, B-III, C-II, D-I  
(D) A-III, B-I, C-IV, D-II

**Q21** The output of the following C Program is:

```
#include <stdio.h>
int main (void)
{
    int x, *p;
    x=30;
    p=x;
    printf ("%d", *p);
    return 0;
}
```

- (A) 30 (B) value of x  
(C) address of x (D) Error

**Q22** Which of the following have truth values if the universe of discourse consists of all integers?

- A.  $\forall n \exists m (n^2 < m)$   
B.  $\exists n \forall m (n < m^2)$   
C.  $\exists n \exists m (n^2 + m^2 = 5)$   
D.  $\exists n \exists m (n^2 + m^2 = 6)$   
E.  $\exists n \exists m (m + n = 4 \wedge n - m = 1)$

Choose the correct answer from the options given below :

- (A) A, B, C Only (B) B, C, E Only  
(C) C, D, E Only (D) B, D Only



**Q23** Multiple Inheritance is permitted directly in....

- A. C++
- B. Java
- C. Python
- D. VB.NET
- E. BASIC

Choose the most appropriate answer from the options given below :

- (A) A and B Only
- (B) A and C Only
- (C) B and D Only
- (D) More than one of the above

**Q24** The ratio of errors found before s/w delivery and to the total errors (before and after) is called :

- (A) Accuracy
- (B) Specificity
- (C) Defect Removal efficiency
- (D) Through put

**Q25** Match List-I with List-II.

| List - I        | List - II                 |
|-----------------|---------------------------|
| A. RAID level-2 | I. block-interleaved      |
| B. RAID level-3 | II. error-correcting-code |
| C. RAID level-4 | III. reed-solomon codes   |
| D. RAID level-6 | IV. bit-interleaved       |

Choose the correct answer from the options given below :

- (A) A-I, B-III, C-IV, D-II
- (B) A-II, B-IV, C-I, D-III
- (C) A-IV, B-I, C-II, D-III
- (D) A-II, B-III, C-IV, D-I

**Q26** Which of the followings sequence  $a_n$  is the solution of recurrence relation  $a_n = 8a_{n-1} - 16a_{n-2}$ ?

- A.  $a_n = 1$
- B.  $a_n = 2^n$
- C.  $a_n = 4^n$
- D.  $a_n = n \cdot 4^n$
- E.  $a_n = (-4)^n$

Choose the most appropriate answer from the options given below :

- (A) A, B, C Only
- (B) B, C, D Only
- (C) C, D Only
- (D) C, D, E Only

**Q27** Match List-I with List-II :

| List - I        | List - II                                                     |
|-----------------|---------------------------------------------------------------|
| A. Thread spawn | I. to convert a blocking system call into a nonblocking call. |
| B. Block thread | II. to reclaim pages                                          |
| C. Jacketing    | III. processor turn to the execution of another ready thread. |
| D. Flush        | IV. thread is provided with its own register context.         |

Choose the correct answer from the options given below :

- (A) A-I, B-III, C-IV, D-II
- (B) A-IV, B-III, C-II, D-I
- (C) A-IV, B-III, C-I, D-II
- (D) A-II, B-I, C-IV, D-III

**Q28** Which of the following is the Markup language?

- A. HTML
- B. XML
- C. DHTML
- D. LML
- E. PML

Choose the correct answer from the options given below :

- (A) B, C, D Only
- (B) A, C, D Only
- (C) A, C, D, E Only
- (D) A, B, C Only

**Q29** Which of the followings are type of interface in an ATM network?

- A. user-to-user
- B. user-to-network
- C. network-to-network
- D. host-to-network
- E. user-to-host

Choose the correct answer from the options given below :

- (A) A and B Only
- (B) B and C Only
- (C) C and D Only
- (D) D and E Only

**Q30** For the fuzzy set  $A =$

$\{(x_1, 0.3), (x_2, 0.7), (x_3, 0.4)\}$  the



complement of A would be represented as :

- (A)  $\{(x_1, 0.4), (x_2, 0.3), (x_3, 0.1)\}$   
 (B)  $\{(x_1, 0.3), (x_2, 0.7), (x_3, 0.4)\}$   
 (C)  $\{(x_1, 0.7), (x_2, 0.3), (x_3, 0.6)\}$   
 (D)  $\{(x_1, 0.21), (x_2, 0.28), (x_3, 0.12)\}$

**Q31** The instructions

ADD R<sub>1</sub>, A, B

ADD R<sub>2</sub>, C, D

MUL X, R<sub>1</sub>, R<sub>2</sub>

evaluates the  $X=(A+B)*(C+D)$  in which of the following?

- (A) 3 - Address instructions  
 (B) 2 - Address instructions  
 (C) 1 - Address instructions  
 (D) RISC instructions

**Q32** An ambiguous grammar is one which has :

- A. More than one derivations  
 B. More than one left most derivations  
 C. More than one right most derivations  
 D. More than one Parse tree  
 E. More than one syntax tree

Choose the correct answer from the options given below :

- (A) A and D Only (B) B and C Only  
 (C) D and E Only (D) A and E Only

**Q33** Back propagation is a learning technique that adjusts weights in the neural network by propagating weight changes :

- (A) Forward from source to hidden nodes  
 (B) Backward from sink to source  
 (C) Forward from source to sink  
 (D) Backward from sink to hidden nodes

**Q34** The Boolean expression for the following would be :

$$F(A, B, C) = \sum(0, 2, 6)$$

$$d(A, B, C) = \sum(1, 3, 5)$$

- (A)  $A'C' + BC'$  (B)  $B+AC'$   
 (C)  $A+BC'$  (D)  $A'+BC'$

**Q35** Arrange the following in correct order, so that they can follow a proper run time environment.

- A. Programming statement  
 B. Compilation

C. Memory allocation

D. Type of variable

E. Memory deallocation

Choose the correct answer from the options given below :

- (A) A, B, C, D, E (B) A, B, D, C, E  
 (C) B, A, D, C, E (D) C, D, B, E, A

**Q36** A computer system with a cache access time of 100 ns, a main memory access time of 1100 ns, and a hit ratio of 0.9, then the average access time would be :

- (A) 200 ns (B) 190 ns  
 (C) 210 ns (D) 120 ns

**Q37** An IPv4 packet has arrived with the first few hexadecimal digits as shown below. How many hops can this packet travel before being dropped? The data belong to what upper-layer protocol?

( 5400001800100110040601...)<sub>16</sub>

- (A) 01 hops, TCP (B) 02 hops, UDP  
 (C) 03 hops, ICMP (D) 4 hops, TCP

**Q38** Arrange the following simple graphs based on a number of spanning trees they have :

- A.  $K_3$  [Complete Graph with 3 nodes]  
 B.  $K_4$  [Complete Graph with 4 nodes]  
 C.  $K_{2,2}$  (Complete bipartite graph of the 2 nodes in each partition)  
 D.  $C_5$  (Cycle graph of 5 nodes)

Choose the correct answer from the options given below :

- (A) A, B, C, D  
 (B) A, C, B, D  
 (C) A, C, D, B  
 (D) D, B, C, A

**Q39** Match List-I with List-II :

| List-I                     | List-II          |
|----------------------------|------------------|
| A. Channelization protocol | I. ALOHA         |
| B. Bit-oriented protocol   | II. TDMA         |
| C. Random-access protocol  | III. Reservation |



|                               |          |
|-------------------------------|----------|
| D. Controlled-access protocol | IV. HDLC |
|-------------------------------|----------|

Choose the correct answer from the options given below:

- (A) A-I, B-III, C-IV, D-II  
 (B) A-II, B-IV, C-I, D-III  
 (C) A-IV, B-III, C-II, D-I  
 (D) A-III, B-IV, C-I, D-II

**Q40** What is the total delay (latency) for a frame size of 10 million bits that is being set up on a link with 20 routers, each having a queuing time of  $2 \mu s$  and a processing time of  $1 \mu s$ ? The length of link 5000 km. The speed of light inside the link is  $2 \times 10^8$  m/s. The link has bandwidth of 6 Mbps.

- (A) 0.624050 s (B) 1.691726 s  
 (C) 2.425080 s (D) 1.714030 s

**Q41** How much time does an algorithm using  $2^{50}$  bit operations need if each bit operation takes  $2^{-38}$  seconds of time?

- (A) 1 hour (B) 10 minutes  
 (C) 30 minutes (D) 1.5 hour

**Q42** Match List - I with List - II.

| List-I (Boolean Expression)          | List-II (Dual of Boolean Expression)       |
|--------------------------------------|--------------------------------------------|
| A. $x \cdot y + 0$                   | I. $(\bar{x} + 0) \cdot (\bar{y} \cdot z)$ |
| B. $\bar{x} \cdot 1 + (\bar{y} + z)$ | II. $x + y \cdot 1$                        |
| C. $\bar{x} \cdot (\bar{y} + 0)$     | III. $\bar{x} + \bar{y} \cdot 1$           |
| D. $x \cdot 1 + (y + \bar{z})$       | IV. $(x + 0) \cdot (y \cdot \bar{z})$      |

Choose the correct answer from the options given below :

- (A) A-III, B-IV, C-II, D-I  
 (B) A-III, B-IV, C-I, D-II  
 (C) A-II, B-I, C-III, D-IV  
 (D) A-II, B-I, C-IV, D-III

**Q43** A bit string of length four is generated at random. What is probability that it contains at least two consecutive 0's given that first bit is 0?

- (A)  $\frac{1}{2}$  (B)  $\frac{5}{16}$   
 (C)  $\frac{5}{8}$  (D)  $\frac{1}{4}$

**Q44** Services provided by point-to-point protocol (PPP):

- A. frame format  
 B. establishment of the link  
 C. flow control  
 D. sequence numbering  
 E. authentication

Choose the correct answer from the options given below

- (A) A, B, and E Only  
 (B) A, C and D Only  
 (C) D, E and B Only  
 (D) B, C and D Only

**Q45** Which of the following is/are true in case of tree locking?

- A. It ensures shorter waiting times.  
 B. It ensures freedom from deadlock.  
 C. Transaction can lock data items earlier.  
 D. It does not ensure greater amount of concurrency.

Choose the correct answer from the options given below :

- (A) A and B Only (B) C and D Only  
 (C) B and C Only (D) A and D Only

**Q46** Arrange the following sets in increasing order, on the basis of their cardinality :

- A.  $A_1 = \{\{1, 2\}, \{3\}\}$   
 B.  $A_2 = \{\{1\}, \{2\}, \{3\}, \{4\}\}$   
 C.  $A_3 = \{\{1, 2, 3, 4, 5, 6\}\}$   
 D.  $A_4 = \{\{1, 2\}, \{2, 3, 4\}, \{5\}\}$   
 E.  $A_5 = \{\{1\}, \{2\}, \{3\}, \{4\}, \{5\}\}$

Choose the correct answer from the options given below :

- (A) A, B, D, E, C (B) A, B, E, D, C  
 (C) A, C, D, B, E (D) C, A, D, B, E

**Q47** The Cardinality of a fuzzy set is :

- (A) 0 (B) finite  
 (C) infinite (D) not known

**Q48** Which of the following is not the goal of reverse Engineering?

- (A) Cope with Complexity  
 (B) Recover lost information  
 (C) Detect side effect  
 (D) Data flow



**Q49** The chairs of an auditorium are to be labelled with a letter followed by a positive two digits integer where labelling is starting with A01. How many maximum possible chairs could be labelled in this way?

- (A) 2600 (B) 2574  
(C) 2340 (D) 2366

**Q50** Arrange in appropriate order, the construction of a finite automata :

- A. Minimum State DFA  
B. Regular Expression  
C. NFA- $\epsilon$   
D. Problem Statement  
E. DFA

Choose the correct answer from the options given below:

- (A) B, C, E, A, D (B) D, B, C, E, A  
(C) C, E, B, A, D (D) A, D, C, B, E

**Q51** A lower-triangular sparse matrix is one :

- (A) in which all the non-zero elements lie only on the leading diagonal.  
(B) in which all the non-zero elements lie above leading diagonal.  
(C) in which all the non-zero elements lie below leading diagonal.  
(D) which is not defined for sparse matrix.

**Q52** The following code :

```
stmt → if expr then stmt else stmt
| if expr then stmt
suffers from :
```

- (A) Ambiguity (B) Left factoring  
(C) Left Recursion (D)  $\lambda$  moves

**Q53** Which of the following is not correct about the virtual memory segmentation ?

- (A) It is not necessary to load all of the segments of a process.  
(B) It has no internal fragmentation.  
(C) It has large virtual address space.  
(D) It provides lower degree of multiprogramming.

**Q54** Match List-I with List-II :

| List - I                   | List - II                  |
|----------------------------|----------------------------|
| A. Union                   | I. Virtual Reality         |
| B. Function                | II. Shadow mask            |
| C. Interactive Environment | III. Subroutine            |
| D. Output device           | IV. User-defined data type |

Choose the correct answer from the options given below :

- (A) A-IV, B-III, C-I, D-II  
(B) A-II, B-IV, C-I, D-III  
(C) A-IV, B-II, C-III, D-I  
(D) A-I, B-III, C-IV, D-II

**Q55** Z is set of integers,  $\langle Z, * \rangle$  forms group if \* in :

- (A)  $*(a, b) = a + b + 2$   
(B)  $*(a, b) = 2a + b$   
(C)  $*(a, b) = (a + b) / 2$   
(D)  $*(a, b) = a^2 + b$

**Q56** Which of the following statements is/are false?

- A. A user can assign privileges to any user on any relation.  
B. A user can assign privileges to any user on its own relations only.  
C. System error that leads to accessibility to the portion of database is an example of malicious attack.  
D. Passwords itself need to be secured from unauthorized access.

Choose the correct answer from the options given below :

- (A) A and C Only  
(B) A and B Only  
(C) B and D Only  
(D) B and C Only

**Q57** Which of the following is not a characteristics of a specialized embedded OS ?

- (A) real time scheduling policy.  
(B) responds to external interrupts  
(C) provides special non-sequential files.  
(D) provides fixed or variable sized partitions.

**Q58** What will be the order of the following fields in a frame relay 'frame format'?



- A. Flag
- B. FCS
- C. address
- D. information

Choose the correct answer from the options given below :

- (A) A, C, D, B
- (B) D, A, C, D
- (C) C, B, A, D
- (D) A, B, D, C

**Q59** The techniques used to handle the phantom problem are

- A. Time stamping
- B. Index locking
- C. Predicate locking
- D. Execution indexing

Choose the correct answer from the options given below :

- (A) A and B Only
- (B) B and C Only
- (C) A and D Only
- (D) C and D Only

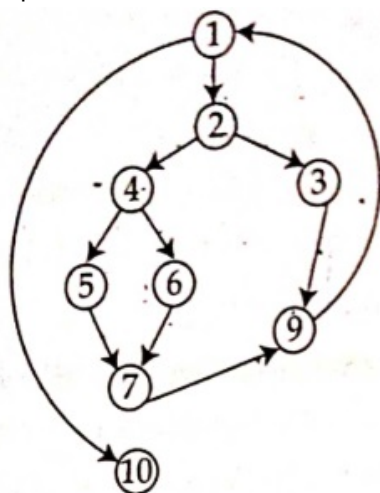
**Q60** The sequence of fields in an IP datagram will be :

- A. flags
- B. options
- C. Protocol
- D. Identification
- E. Service Type

Choose the correct answer from the options given below :

- (A) E, D, A, C, B
- (B) D, A, B, C, E
- (C) D, E, C, B, A
- (D) E, C, D, A, B

**Q61** The cyclomatic complexity of the following flow graph is :



(A) 3

(B) 4

(C) 10

(D) 2

**Q62** IP is responsible for \_\_\_\_\_ communication while TCP is responsible for \_\_\_\_\_ communication.

- (A) Node-to-node, Host-to-host
- (B) Process-to-process, Host-to-host
- (C) Socket-to-socket, Host-to-node
- (D) Host-to-Host, Process-to-process

**Q63** Arrange the steps of Mathematical Modeling :

- A. Solution fitting with data
- B. Formulation Mathematically
- C. Solve Mathematically
- D. Repeat formulation if it fits worst with the data
- E. Interpretation results

Choose the correct answer from the options given below :

- (A) B, C, A, D, E
- (B) B, A, D, C, E
- (C) D, A, B, C, E
- (D) E, D, C, A, B

**Q64** In windows scheduling, which of the following option is correct ?

- (A) 4 non-real-time priorities
- (B) 8 non-real-time priorities
- (C) 12 non-real-time priorities
- (D) 16 non-real-time priorities

**Q65** The I/O methods in which memory addresses and I/O addresses are distinct is called

- A. Isolated I/O
- B. Memory-Mapped I/O
- C. Strobe Control
- D. Handshaking
- E. Interrupt

Choose the most appropriate answer from the options given below :

- (A) A, B Only
- (B) C, D Only
- (C) C, E Only
- (D) B, E Only

**Q66** What will be the ordering of typical steps when processing a high-level query?

- A. Query optimizer
- B. Runtime database processor
- C. Scanning, passing and validating
- D. Query code generator



Choose the correct answer from the options given below :

- (A) D, A, C, B                      (B) C, A, D, B  
(C) C, D, B, A                      (D) D, C, B, A

**Q67** Which of the followings are tautology ?

- A.  $(p \vee r) \rightarrow p$   
B.  $p \rightarrow (p \wedge r)$   
C.  $\sim p \rightarrow (p \rightarrow r)$   
D.  $(p \wedge r) \rightarrow (p \rightarrow r)$   
E.  $\sim(p \rightarrow r) \rightarrow p$

Choose the correct answer from the options given below :

- (A) A, B Only  
(B) A, B, C Only  
(C) B, C, D Only  
(D) C, D, E Only

**Q68** Arrange the following from fastest to slowest in speed.

- A. DRAM  
B. SRAM  
C. CPU Registers  
D. Magnetic tapes  
E. Hard disk

Choose the correct answer from the options given below :

- (A) A, B, C, D, E                      (B) C, B, A, D, E  
(C) C, D, B, A, E                      (D) B, C, D, A, E

**Q69** The language of the following PDA would be :

- $\delta(q_0, a, z_0) = (q_0, aaa z_0)$   
 $\delta(q_0, a, a) = (q_0, aaaa)$   
 $\delta(q_0, b, a) = (q_1, \lambda)$   
 $\delta(q_1, b, a) = (q_1, \lambda)$   
 $\delta(q_1, \epsilon, z_0) = (q_f, z_0)$ , where  $q_f$  is the final state.  
(A)  $\{a^n b^n \mid n > 0\}$                       (B)  $\{a^n b^{2n} \mid n > 0\}$   
(C)  $\{a^n b^{3n} \mid n > 0\}$                       (D)  $\{a^n b^{4n} \mid n > 0\}$

**Q70** Regarding the subroutines, see the following sequence of tasks to be performed :

- A. Turn on the interrupt facility on.  
B. Service the device whose flag is set.  
C. Save the contents of processor registers.

D. Check which flag is set.

E. Restore the contents of processor registers.

Choose the correct answer from the options given below :

- (A) A, D, C, B, E                      (B) D, A, B, C, E  
(C) C, D, B, E, A                      (D) B, D, A, C, E

**Q71** Match List-I with List-II :

| List-I                           | List-II               |
|----------------------------------|-----------------------|
| A. Verification                  | I. White box Testing  |
| B. Validation                    | II. Black box Testing |
| C. Internal logic exercise       | III. Quality Control  |
| D. Software requirement exercise | IV. Quality Assurance |

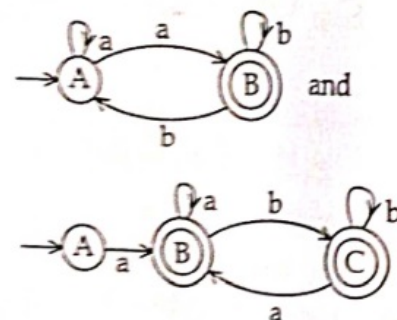
Choose the correct answer from the options given below :

- (A) A-IV, B-III, C-I, D-II  
(B) A-I, B-II, C-III, D-IV  
(C) A-IV, B-III, C-II, D-I  
(D) A-III, B-IV, C-I, D-II

**Q72** The parameter Actual count is used in Genetic Algorithm (GA) for :

- (A) Crossover  
(B) Mutation  
(C) Selecting population  
(D) Encoding the Genetic Algorithm

**Q73** For the given two machines which of the following is correct?



- (A) both Machines are DFAs  
(B) both are non-equivalent Machines  
(C) both are equivalent Machines  
(D) cannot be determined

**Q74**



The number of tokens in the following line : ' for i in range (begin; end ; step) :' is:

- (A) 4 (B) 7  
(C) 8 (D) 12

**Q75** Which of the following structures of knowledge representation are the most purely syntactic?

- A. Conceptual dependency  
B. predicate logic  
C. slot-and-filler  
D. nonmonotonic systems

Choose the correct answer from the options given below :

- (A) A and B Only (B) B and C Only  
(C) B and D Only (D) C and D Only

**Q76** In Genetic Algorithm's cross over operation, mask is used in ?

- (A) Three parent crossover  
(B) Two parent crossover  
(C) Uniform crossover  
(D) N point crossover

**Q77** Match the List-I and List-II

| List-I          | List-II                             |
|-----------------|-------------------------------------|
| A. Complement   | I. Adder                            |
| B. E-OR         | II. Instruction decoding            |
| C. Accumulator  | III. Negative number representation |
| D. Control Unit | IV. Arithmetic and Logic operations |

Choose the correct answer from the options given below :

- (A) A-I, B-II, C-III, D-IV  
(B) A-III, B-II, C-I, D-IV  
(C) A-III, B-I, C-IV, D-II  
(D) A-IV, B-II, C-III, D-I

**Q78** Arrange the following prefix expressions based on their values in increasing order :

- A.  $+ * 2 / 8432$   
B.  $\uparrow * 33 * 425$   
C.  $+ \uparrow 32 \uparrow 23 / 6-42$   
D.  $* +3+3 \uparrow 3+113$

Choose the correct answer from the options given below :

- (A) A, B, C, D (B) B, A, C, D  
(C) D, C, B, A (D) C, D, A, B

**Q79** Which of the following is not the most common property in social network ?

- (A) Degree distribution follows scale free.  
(B) Average shortest path lengths are shorten in connected network.  
(C) Most of them are directed network.  
(D) Follows small world property.

**Q80** For line drawing, which of the following algorithms are used ?

- A. DDA algorithm  
B. Bresenham algorithm  
C. Mid point algorithm  
D. Curve algorithm

Choose the most appropriate answer from the options given below :

- (A) B and C Only  
(B) C and D Only  
(C) A and B Only  
(D) D and A Only

**Q81** According to the Nyquist theorem to produce the original analog signal, one necessary condition is :

- (A) Sampling rate must be at least 2 times the highest frequency contained in the signal  
(B) Sampling rate must be at least 2 times the lowest frequency contained in the signal  
(C) Sampling rate must be at least  $\frac{1}{2}$  times the highest frequency contained in the signal  
(D) Sampling rate must be at least  $\frac{1}{2}$  times the lowest frequency contained in the signal

**Q82** For the fuzzy sets

- $A = \{(x_1, 0.3), (x_2, 0.7), (x_3, 0.4)\}$  and  
 $B = \{(x_1, 0.5), (x_2, 0.2), (x_3, 0.5)\}$ , then  
 $A \cap B$  would be:  
(A)  $\{(x_1, 0.3), (x_2, 0.7), (x_3, 0.4)\}$   
(B)  $\{(x_1, 0.5), (x_2, 0.2), (x_3, 0.5)\}$   
(C)  $\{(x_1, 0.5), (x_2, 0.7), (x_3, 0.5)\}$   
(D)  $\{(x_1, 0.3), (x_2, 0.2), (x_3, 0.4)\}$



- Q83** The least values of a signed and unsigned data type variables requiring 8 bits, would be...
- (A) 0,128                      (B) -128,127  
(C) 0,0                         (D) -128,0
- Q84** Which of the following are true for a complete binary tree?
- A. It has always odd number of vertices.  
B. With  $i$  internal vertices, it has  $i+1$  leaves.  
C. With  $l$  leaves, it has  $2l-1$  vertices.  
D. With  $2n-1$  vertices, it has  $n$  leaves.
- Choose the correct answer from the options given below :
- (A) A, B, C Only  
(B) B, C Only  
(C) A, D Only  
(D) A, B, C, D
- Q85** For making a memory of size  $2048 \times 8$  bytes, by using the chips of size  $128 \times 4$  bytes, the number of chips required is :
- (A) 8                              (B) 16  
(C) 32                            (D) 64
- Q86** Which of the following are sequential circuits?
- A. Flip flop  
B. Multiplexer  
C. Counter  
D. Decoder  
E. Adder
- Choose the correct answer from the options given below :
- (A) B, D Only  
(B) A, C Only  
(C) C, E Only  
(D) D, A Only
- Q87** Which of the following are White box Testing?
- A. Path Testing  
B. Loop Testing  
C. Regression Testing  
D. Non-functional Testing
- Choose the correct answer from the options given below :
- (A) A and B Only              (B) B and C Only  
(C) C and D Only              (D) A and D Only

- Q88** Techniques for performing I/O :
- A. Programmed I/O  
B. Interrupt-driven I/O  
C. Rotational I/O  
D. Direct memory access  
E. Channelized I/O

Choose the correct answer from the options given below :

- (A) A and C Only  
(B) A, B, and D Only  
(C) A, D and E Only  
(D) B and E Only

- Q89** For the software quality assurance, which of the following is correct?

- A. Execution of code  
B. No execution of code  
C. Preventive technique  
D. A corrective technique

Choose the correct answer from the options given below :

- (A) A, B Only                      (B) B, C Only  
(C) C, D only                      (D) D, A Only

- Q90** Which of the following genetic algorithm operations is computationally most expensive?

- (A) Convergence testing  
(B) Initial population creation  
(C) Selection of sub-population for mating  
(D) Reproduction to produce next generation

**Directions (91-95) Read the following passage and answer the given questions.**

Answer the following 5 questions based on the passage, (141-145)

A firm can produce two types of cloth say A and B. Three kinds of wool are required for it, say red wool, green wool, and blue wool. One unit length of type A cloth needs 2 yards of red and 3 yards of green wool. One unit length of type B cloth needs 3 yards of red wool, 2 yards of green wool and 2 yards of blue wool. The firm has only a stock of 8 yards of red wool, 10 yards of green wool, and 15 yards of blue wool. It is assumed that the income obtained from one unit length of



type A and type B clothes are ₹ 30 and ₹ 50, respectively.

**Q91** When we formulate the LPP from the above-given passage. Which of the following is an objective function for its dual problem?

- (A)  $\text{Min } P = 5u + 8v + 15w$
- (B)  $\text{Min } P = 8u + 10v + 15w$
- (C)  $\text{Min } P = 15u + 5v + 8w$
- (D)  $\text{Max } P = 10u + 8v + 15w$

**Q92** Which of the following will be appropriate objectives function for the given problem ?

- (A)  $2x+3y$                       (B)  $3x+2y+2z$
- (C)  $30x+50y$                 (D)  $3x+2y$

**Q93** How many unit length used for cloth A when income is maximized ?

- (A) 0.8 unit                      (B) 2.4 unit
- (C) 2.8 unit                      (D) 3.0 unit

**Q94** If  $x$  unit length of type A cloths and  $y$  unit length of type B cloths, then constrains formulate for the LPP, is :

- (A)  $y \leq 7.5, 2x + 3y \leq 8, 3y + 2x \leq 10$
- (B)  $y \leq 7.5, 2x + 3y \leq 8, 3x + 2y \leq 10$
- (C)  $2x + 3y \leq 8, 3x + 2y + 2z < 15$
- (D)  $2x + 3y \leq 10, 3x + 2y \leq 10, 2y \leq 15$

**Q95** On the basis of available material, maximum total income is :

- (A) ₹ 133.3                      (B) ₹ 124
- (C) ₹ 100                        (D) ₹ 153.3

**Directions (96-100) Read the following passage and answer the given questions.**

A company is implementing a new database system to manage its customer information. The database contains labels for customers, orders, products and sales representatives. Each customer can place multiple orders and each

order can contain multiple products. Each order is associated with a specific sales representative who manages the accountable products. The database is designed to improve efficiency in tracking sales and customer interactions.

**Q96** Which SQL query retrieves the total number of orders placed by a specific customer ?

- (A) `Select COUNT (*) As total_orders from orders where customer_id = [Customer_ID];`
- (B) `Select COUNT (order_id) As total_orders from orders where customer_id = [Customer_ID];`
- (C) `Select COUNT (DISTINCT order_id) As total_orders from orders Where customer_id =[ Customer_ID];`
- (D) `Select SUM (order_Id) As total_orders from orders where customer_id = [Customer_ID];`

**Q97** Based on the passage, which of the following statements is true ?

- (A) Each customer can place only one order.
- (B) Each sales representative can manage multiple customer accounts.
- (C) Each order can be associated with only one product.
- (D) The database does not include a table for products.

**Q98** Which of the following tables is likely to have a foreign key referencing another table in the database system implemented by the company ?

- (A) Customers
- (B) Orders
- (C) Products
- (D) Sales representatives

**Q99** Which SQL query retrieves the total number of distinct customers who have placed orders?

- (A)



- Select COUNT (DISTINCT customer\_id)  
As total\_customers From orders;
- (B) Select COUNT (\*) As total\_customers  
from customers;
- (C) Select COUNT (DISTINCT order\_id) As  
total\_customers from orders;
- (D) Select COUNT (DISTINCT customer\_id)  
As total\_customers from customers;

- Q100** Which of the following best describes the relationship between the "orders" table and the "Customers" table in the company's database system?
- (A) One-to-One      (B) One-to-Many  
(C) Many-to-One    (D) Many-to-Many



## Answer Key

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Q1 (D)  
Q2 (D)  
Q3 (B)  
Q4 (B)  
Q5 (A)  
Q6 (D)  
Q7 (A)  
Q8 (C)  
Q9 (B)  
Q10 (A)  
Q11 (C)  
Q12 (B)  
Q13 (A)  
Q14 (D)  
Q15 (B)  
Q16 (C)  
Q17 (B)  
Q18 (A)  
Q19 (C)  
Q20 (D)  
Q21 (D)  
Q22 (A)  
Q23 (B)  
Q24 (C)  
Q25 (B)  
Q26 (C)  
Q27 (C)  
Q28 (D)  
Q29 (B)  
Q30 (C)  
Q31 (A)

Q32 (C)  
Q33 (B)  
Q34 (D)  
Q35 (B)  
Q36 (A)  
Q37 (D)  
Q38 (C)  
Q39 (B)  
Q40 (B)  
Q41 (A)  
Q42 (C)  
Q43 (C)  
Q44 (A)  
Q45 (A)  
Q46 (D)  
Q47 (B)  
Q48 (D)  
Q49 (B)  
Q50 (B)  
Q51 (C)  
Q52 (A)  
Q53 (D)  
Q54 (A)  
Q55 (A)  
Q56 (A)  
Q57 (C)  
Q58 (A)  
Q59 (B)  
Q60 (A)  
Q61 (B)  
Q62 (D)



Q63 (A)  
Q64 (D)  
Q65 (A)  
Q66 (B)  
Q67 (D)  
Q68 (B)  
Q69 (C)  
Q70 (C)  
Q71 (A)  
Q72 (C)  
Q73 (B)  
Q74 (D)  
Q75 (B)  
Q76 (C)  
Q77 (C)  
Q78 (B)  
Q79 (B)  
Q80 (C)  
Q81 (A)

Q82 (D)  
Q83 (D)  
Q84 (D)  
Q85 (C)  
Q86 (B)  
Q87 (A)  
Q88 (B)  
Q89 (B)  
Q90 (D)  
Q91 (B)  
Q92 (C)  
Q93 (C)  
Q94 (B)  
Q95 (B)  
Q96 (B)  
Q97 (B)  
Q98 (B)  
Q99 (A)  
Q100 (C)



## Hints & Solutions

### Q1 Text Solution:

Option A) Perl: Perl is a general-purpose, high-level programming language that supports multiple programming paradigms, including procedural, functional, and object-oriented programming. It does support OOP concepts, so it is not the correct answer.

Python: Python is a widely-used high-level programming language known for its simplicity and readability. It fully supports object-oriented programming with classes, objects, inheritance, polymorphism, etc. Therefore, Python is an object-oriented programming language.

Smalltalk: Smalltalk is a pure object-oriented programming language that was designed to be simple and easy to use. It heavily emphasizes objects and messages between them. Smalltalk is an object-oriented programming language.

SQL Plus: SQL (Structured Query Language) is a special-purpose language primarily used for managing and manipulating databases. SQL Plus, however, is an interactive tool for Oracle databases that includes a command-line interface. SQL and SQL Plus are not considered object-oriented programming languages as they do not support the creation of objects, classes, or inheritance.

So option D is the correct answer.

### Q2 Text Solution:

Shift-reduce conflicts occur in parser generators (like in parser combinators or parser generators for languages like LR parsers) when the parser encounters a situation where it can either shift (move to the next state and expect more input) or reduce (apply a production rule and reduce the current stack of symbols).

To resolve shift-reduce conflicts, parser generators typically use the following mechanisms:

Option D) Associativity and Precedence: This is the most common method to resolve shift-reduce conflicts. Operators in programming

languages often have associativity (whether operations are evaluated left-to-right or right-to-left) and precedence (which operation takes precedence over another). By specifying the associativity and precedence of operators in the grammar, the parser can decide whether to shift or reduce based on the current operator and the one at the top of the stack.

Associativity: Determines the direction (left or right) in which operations of the same precedence are evaluated. For example, in many programming languages, addition and subtraction are left-associative.

Precedence: Determines the order of operations when multiple operators are present. For example, multiplication has higher precedence than addition.

Option A) Left Recursion: Left recursion is a parsing technique where the left-most non-terminal symbol in a production rule appears again as the first symbol in the right-hand side of the rule. Left recursion can sometimes lead to conflicts, but it is not typically used to resolve shift-reduce conflicts directly.

Option B) Left Factoring: Left factoring is a technique used to refactor grammar rules to eliminate common prefixes. This technique can help in reducing the number of conflicts but is not directly used to resolve shift-reduce conflicts.

Option C) Ambiguity: Ambiguity in grammar can lead to conflicts, including shift-reduce conflicts. Resolving ambiguity involves restructuring the grammar to remove multiple interpretations of the same input string, but this does not directly resolve shift-reduce conflicts.

So option D is the correct answer.

### Q3 Text Solution:

To determine the least integer  $n$  such that  $f(x)$  is  $O(x^n)$  for each function  $f(x)$ , let's analyze each function given:

A.  $f(x) = 3x^3 + (\log x)^4$



For large  $x$ ,  $(\log x)^4$  grows much slower compared to  $x^3$

Therefore,  $f(x)$  is asymptotically dominated by  $x^3$

Hence,

$$f(x) = O(x^3)$$

$$B. f(x) = (x^5 + x^2 + 1) / (x^3 + 1)$$

As  $x \rightarrow \infty$ , the dominant term in the numerator and the denominator is  $x^5$  and  $x^3$ , respectively.

$$\text{Hence, } f(x) \sim x^2$$

$$\text{Thus, } f(x) = O(x^2)$$

$$C. f(x) = (x^3 + x^2 + 1) / (x^2 + 1)$$

As  $x \rightarrow \infty$ , the dominant term in the numerator and the denominator is  $x^3$  and  $x^2$ , respectively.

$$\text{Hence, } f(x) \sim x.$$

$$\text{Thus, } f(x) = O(x).$$

$$D. f(x) = (x^2 + 5 \log x) / (x^2 + 1)$$

For large  $x$ ,  $(\log x)$  grows much slower compared to  $x^2$

The dominant terms are  $x^2$  in both numerator and denominator.

$$\text{Therefore, } f(x) \sim 1.$$

$$\text{Thus, } f(x) = O(1)$$

So, the correct order of increasing  $n$  is D, C, B, A.

So option B is the correct answer.

#### Q4 Text Solution:

The Software process framework is required for representing common process activities. Five framework activities are described in a process framework for software engineering.

Communication: By communication, customer requirement gathering is done. Communication with consumers and stakeholders to determine the system's objectives and the software's requirements.

Planning: Establish engineering work plan, describe technical risk, list resources requirements, work produced, and define work schedule.

Modeling: Architectural models and design to better understand the problem and to work towards the best solution.

Construction: Creating code, testing the system, fixing bugs, and confirming that all criteria are

met.

Deployment: In this activity, a complete or non-complete product or software is represented to the customers to evaluate and give feedback. On the basis of their feedback, we modify the product for the supply of better products. So option B is the correct answer.

#### Q5 Text Solution:

$$A. A \rightarrow aB|a:$$

This form suggests productions where A can derive either a terminal symbol followed by a non-terminal symbol or just a terminal symbol itself. This is typical of regular grammars or regular expressions, which can be recognized by finite automata.

Matches with IV. Finite Automaton: Regular grammars are equivalent to regular languages, which can be recognized by finite automata.

$$B. A \rightarrow BC|a:$$

This form suggests productions where A can derive either a combination of two non-terminal symbols or a single terminal symbol. This form is characteristic of Chomsky Normal Form.

$$C. LL(1) \text{ grammar:}$$

An LL(1) grammar is a type of context-free grammar that can be parsed by a recursive descent parser without backtracking.

$$D. \text{ Halting problem:}$$

The halting problem is undecidable for Turing machines.

So option A is the correct answer.

#### Q6 Text Solution:

Option A) A double-layer auto-associative neural network: This option describes a specific type of neural network different from a perceptron. Auto-associative neural networks are used for pattern completion and pattern recognition tasks, and they are not the same as a perceptron.

Option B) A neural network that contains feedback: This describes a recurrent neural network (RNN), not a perceptron. RNNs have feedback loops that allow them to maintain information in memory, unlike perceptrons.



Option C) An auto-associative neural network: This option refers to a type of neural network where the network is trained to generate an output that is similar to its input. It's not specifically describing a perceptron.

Option D) A single-layer feed-forward neural network with pre-processing: This accurately describes a perceptron. Perceptrons are single-layer feedforward neural networks without hidden layers, and they process inputs through a weighted sum with an activation function.

So option D is the correct answer.

#### Q7 Text Solution:

Unlike rule-based systems and machine learning approaches, neural networks use complex algorithms that can learn from massive data sets to generate responses that are more natural-sounding and contextually relevant.

A. Spam email detection using rule-based systems:

Relies on predefined rules and pattern matching. The least complex among the given options.

B. Handwritten digit recognition using shallow neural networks. A shallow neural network has only one (or just a few) hidden layers between the input and output layers. The input layer receives the data, the hidden layer(s) process it, and the final layer produces the output. More complex than rule-based systems but simpler compared to CNNs and reinforcement learning.

C. Image classification using convolutional neural networks. These neural networks facilitate superior performance with image, speech, or audio signal inputs. They have three main types of layers, which are:

- Convolutional layer
- Pooling layer
- Fully-connected (FC) layer

D. Autonomous driving using reinforcement learning algorithms.

Reinforcement learning involves training agents to make sequential decisions in an environment

to maximize cumulative rewards.

Autonomous driving using reinforcement learning is highly complex due to the real-time decision-making, environment modeling, and safety considerations involved.

So option A is the correct answer.

#### Q8 Text Solution:

A. Line coding scheme: This refers to the different coding techniques used in digital communication to represent binary data on a transmission line.

I. NRZ-L: NRZ-L (Non-Return-to-Zero Level) has a bandwidth calculated as  $B=N/2$ .

Matches with IV.

II. 2B1Q: 2B1Q (2 Binary 1 Quaternary). It can send more than one data with signal and hence saves the bandwidth required. Required average BW equals  $N/4$ .

Matches with I.

III. 4D-PAM 5: 4D-PAM 5 (4-Level Pulse Amplitude Modulation with 5 levels) has an average bandwidth calculated as  $B=N/8$ .

Matches with II.

IV. MLT-3: MLT-3 (Multi-Level Transmit 3) has an average bandwidth calculated as  $B=N/3$ .

Matches with III.

So option C is the correct answer.

#### Q9 Text Solution:

Steps of Address Translation in Linux Virtual Memory Scheme-

Register: The virtual address generated by the CPU starts with the CPU register, which contains the virtual address.

First-Level Page Table (Page Directory): This table resides in physical memory at all times. It acts as a high-level index, containing entries that point to the second-level page tables known as page middle Directory. Second-Level Page Tables:

These tables hold PTEs (Page Table Entries) for a smaller chunk of the virtual address space.

Offset is a fixed value within a virtual address that remains constant and identifies the specific location of the data within a virtual page.

So option B is the correct answer.

#### Q10 Text Solution:



If  $f(n) = 5f(n/2) + 3$  and  $f(1) = 7$

Then

$$f\left(\frac{n}{2}\right) = 5f\left(\frac{n}{4}\right) + 3$$

$$f(n) = 5\left(5f\left(\frac{n}{4}\right) + 3\right) + 3$$

$$f(n) = 5^2 f\left(\frac{n}{4}\right) + 5 \cdot 3 + 3$$

$$f(n) = 5^2 f\left(\frac{n}{4}\right) + 15 + 3$$

$$f(n) = 5^k f\left(\frac{n}{2^k}\right) + 3(5^0 + 5^1 + \dots + 5^{k-1})$$

$$f(n) = 5^k f\left(\frac{n}{2^k}\right) + 3\left(\frac{5^k - 1}{4}\right)$$

now put  $n = 2^k$

$$f(2^k) = 5^k f(1) + 3\left(\frac{5^k - 1}{4}\right)$$

$$= 5^k \cdot 7 + 3\left(\frac{5^k - 1}{4}\right)$$

$$= 5^k \cdot 7 + \frac{3}{4}(5^k - 1)$$

$$= 5^k \cdot 7 + \frac{3 \cdot 5^k}{4} - \frac{3}{4}$$

$$= \frac{5^k \cdot 28 + 5^k \cdot 3 - 3}{4}$$

$$= 5^k \left(\frac{31}{4}\right) - \frac{3}{4}$$

Thus option A is the correct answer.

#### Q11 Text Solution:

Option A) Direct Addressing Mode - In direct addressing mode, the address field in the instruction contains the effective address of the operand.

Example: Add the content of R1 and 1001 and store back to R1:

Add R1, (1001)

Here 1001 is the address where the operand is stored.

Option B) In Indirect addressing mode, the address field in the instruction contains the memory location or register where the effective address of the operand is present.

LOAD R1, @500

Option C) Index Addressing Mode - Index addressing mode involves adding an index register (in this case, X) to a base address (in this case, ADR) to form the effective address.

This matches the expression  $A \leftarrow M[ADR + XR]$ , where ADR acts as the base address and XR as the index.

Option D) Immediate Addressing Mode -

Immediate addressing mode involves specifying the operand directly in the instruction, such as LD #value, where #value is the immediate value to be loaded.

So option C is the correct answer.

#### Q12 Text Solution:

Crossover is a technique used in artificial intelligence, machine learning, and evolutionary computation whereby two or more solutions are combined to create a new solution.

Option A) Two-Point Crossover: This is a specific case of a N-point Crossover technique. Two random points are chosen on the individual chromosomes (strings) and the genetic material is exchanged at these points.

Option B) Single Point Crossover: A point on both parents' chromosomes is picked randomly, and designated a 'crossover point'. Bits to the right of that point are swapped between the two parent chromosomes. This results in two offspring, each carrying some genetic information from both parents

Option C) N point crossover is a genetic operator used in genetic algorithms to generate new solutions by exchanging segments of genetic material (genes or bits) between parents at N predefined points.

Option D) Uniform Crossover: Each gene (bit) is selected randomly from one of the corresponding genes of the parent chromosomes.

So option B is the correct answer.

#### Q13 Text Solution:

The sequence of steps for a page replacement algorithm typically follows a specific order when a page fault occurs and a new page needs to be brought into memory while replacing an existing page

D. No frame is free:



If all frames are currently occupied, the algorithm needs to select a page to evict from memory to make space for the new page.

A. Find a frame that is not currently being used:  
The algorithm checks if there is a free frame in memory that is not currently used and can be allocated for the new page.

E. Free a frame

Various page replacing algorithms are there e.g. FIFO, and LRU etc, based on these we free a frame for the new page.

B. Change the page table

After selecting a frame for the new page, the page table is updated to reflect the new mapping between the page and the frame.

C. use a freed frame

The new page is then loaded into this freed frame.

This step optimizes memory usage by reusing frames that were previously occupied.

So option A is the correct answer.

#### Q14 Text Solution:

A. Binary Search: Binary Search involves dividing the array in half and searching in the appropriate half.

Matches with IV.  $f(n) = f(n/2) + 2$

Explanation: Binary search reduces the problem size by half in each step, and the +2 accounts for constant time operations per step.

B. Find maximum and minimum of a sequence:

This problem can be solved using a divide-and-conquer approach, similar to finding an element in a sorted array.

Matches with III.  $f(n) = 2f(n/2) + 2$

Explanation: In this problem, the sequence is divided into two parts recursively, and then the results from each part are combined. The +2 accounts for the constant time to combine results.

C. Merge Sort:

Merge Sort recursively divides the array into halves, sorts each half, and then merges them.

$f(n) = 2f(n/2) + n$

Explanation: Merge Sort divides the array into

two halves recursively, sorts each half, and merges them. The +n accounts for the linear time to merge two sorted halves.

D. Fast Matrix Multiplication: Algorithms like Strassen's algorithm use a divide-and-conquer approach for fast matrix multiplication.

Matches with I.  $f(n) = 7f(n/2) + 15n^2/4$

Explanation: Strassen's algorithm divides matrices into submatrices and uses matrix operations recursively. The recurrence reflects the complexity of matrix multiplication at each level.

So option D is the correct answer.

#### Q15 Text Solution:

Option A) Mutual exclusion is a program object that blocks multiple users from accessing the same shared variable or data at the same time.

Option B) Data that is coherent is data that is the same across the network. In other words, if data is coherent, data on the server and all the clients is synchronized.

Option C) Starvation occurs when a process in the OS runs out of resources because other processes are using it.

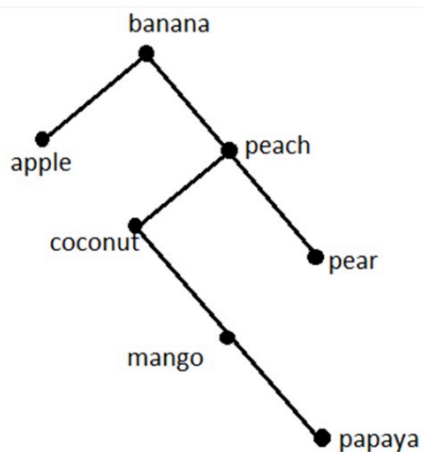
Option D) Deadlock in OS refers to a situation where more than one or two processes or threads are not able to proceed because each is waiting for the other to release a resource.

So when processes are unaware of each other during execution option B data coherent does not exist. Thus option B is the correct answer.

#### Q16 Text Solution:

Let us draw the binary search tree in alphabetical order





From the above diagram we can see the height of binary search tree is 4.

So option C is the correct answer.

#### Q17 Text Solution:

A. Batch Multiprogramming: To speed up processing, jobs with similar needs are batched together and run as a group. The programmers leave their programs with the operator and the operator then sorts the programs with similar requirements into batches.

B. A time-sharing processor's time which is shared among multiple users simultaneously is termed as time-sharing.

The main difference between Multiprogrammed Batch Systems and Time-Sharing Systems is that in the case of Multiprogrammed batch systems, the objective is to maximize processor use, whereas in Time-Sharing Systems, the objective is to minimize response time.

So A matches with III and B matches with IV.

C. In a simple batch-processing scheme, the use of a piece of software known as the monitor. The user submits the job on cards or tape to a computer operator, who batches the jobs together sequentially and places the entire batch on an input device, for use by the monitor where the user no longer has direct access to the processor. So C matches with II.

D. A reentrant procedure is one in which a single copy of the program code can be shared by multiple users during the same period of time. Re-entrance has two key aspects: The program code cannot modify itself and the local data for

each user must be stored separately. In a shared system, reentrancy allows more efficient use of main memory: One copy of the program code is kept in the main memory, but more than one application can call the procedure. So D matches with I.

Thus option B is the correct answer.

#### Q18 Text Solution:

Option A) Type conversion: This is typically handled during the semantic analysis phase, where the compiler checks if types are compatible and converts them if necessary.

Option B) Tokenization: Tokenization is part of the lexical analysis phase (often called scanning), which precedes semantic analysis. It involves breaking down the source code into tokens (keywords, identifiers, literals, etc.) for further processing.

Option C) Loop optimization: This is part of the optimization phase of compilation, where the compiler analyzes loops and optimizes them for performance.

Option D) Data flow Analysis- To efficiently optimize the code compiler collect all the information about the program and distribute this information to each block of the flow graph. This process is known as data-flow graph analysis which is done at code optimization phase.

So option A is the correct answer.

#### Q19 Text Solution:

CFL is not closed under Intersection, complementation and set difference.

| Property        | Regular language | CFL    |
|-----------------|------------------|--------|
| Union           | Closed           | Closed |
| Intersection    | Closed           | No     |
| Set Difference  | Closed           | No     |
| Complementation | Closed           | No     |
| Kleen star      | Closed           | Closed |
| Concatenation   | Closed           | Closed |

So option C is the correct answer.



**Q20 Text Solution:**

- Post Correspondence Problem is a popular undecidable problem that was introduced by Emil Leon Post in 1946. It is simpler than Halting Problem. In this problem we have N number of Dominos (tiles). The aim is to arrange tiles in such order that string made by Numerators is same as string made by Denominators.
- LR parser Also called as Canonical LR parser. Generates large table and large number of states.
- If the DFA has n states, and the language contains any string of length n or more, then the language is infinite.

Predictive parsing is a special form of recursive descent parsing, where no backtracking is required.

**Q21 Text Solution:**

Here  $p=x$ ; is given.

In this line, p is a pointer to an integer (int \*), and x is an integer. However, the assignment  $p = x$ ; attempts to assign an integer (x) to a pointer (p). This will result in a compilation error because we cannot assign an integer directly to a pointer.

Corrected program should be as follows-

```
#include <stdio.h>
```

```
int main(void)
```

```
{
    int x, *p;
    x = 30;
    p = &x;
    printf("%d", *p);
    return 0;
}
```

Explanation of Corrected Program:

- $\text{int } x, *p$ ;; Declares an integer x and a pointer to an integer p.
- $x = 30$ ;; Assigns the value 30 to the integer x.
- $p = \&x$ ;; Assigns the address of x to the pointer p.
- $\text{printf}(\text{"\%d"}, *p)$ ;; Prints the value of x (which is 30) by dereferencing p.

- $\text{return } 0$ ;; Indicates successful completion of  $\text{main}()$ .
- This corrected version of the program will output 30, which is the value stored in x.

So option D is the correct answer.

**Q22 Text Solution:**

Statement A:  $\exists n \exists m (n^2 < m)$

This statement asserts the existence of integers n and m such that  $n^2 < m$ .

For integers, this statement is true because, for example, if  $n=0$  and  $m=2$ , then  $0^2 = 0 < 2$ .

Statement B:  $\exists n \exists m (n < m^2)$

This statement asserts the existence of integers n and m such that  $n < m^2$ .

For integers, this statement is true because, for example, if  $n=0$  and  $m=2$ , then  $0 < 2^2 = 4$ .

Statement C:  $\exists n \exists m (n^2 + m^2 = 5)$

Substitute  $n=1$  and  $m=2$ :  $1^2 + 2^2 = 1 + 4 = 5$  holds true.

Statement D:  $\exists n \exists m (n^2 + m^2 = 6)$

This statement asserts the existence of integers n and m such that  $n^2 + m^2 = 6$ .

For integers, this statement is false because, for example,  $n=2$  and  $m=2$  it does not satisfy  $2^2 + 2^2 = 4 + 4 = 8$ .

Statement E:  $\exists n \exists m ((m+n=4) \wedge (n-m=1))$

This statement asserts the existence of integers n and m such that:  $m+n=4$  and  $n-m=1$

Now substitute  $n=3$  and  $m=1$

$m+n = 1+3 = 4$  but  $3-1 = 2$  which is false. Therefore, we cannot find integers n and m that satisfy both equations simultaneously.

Hence, Statement E  $\exists n \exists m (m+n=4 \wedge n-m=1)$  is false in the universe of discourse of all integers.

So Option A is the correct answer.

**Q23 Text Solution:**

**Explanation:**

**A:** C++ supports multiple inheritance directly.

Multiple inheritance allows a class to inherit from more than one base class. For example:

```
class Base1 {
    // class members
};
```



```
class Base2 {
    // class members
};
class Derived : public Base1, public Base2 {
    // class members
};
```

Here, Derived inherits from both Base1 and Base2.

**B: Java does not support multiple inheritance of classes directly.** However, Java supports multiple inheritance of interfaces, which are abstract classes by nature. A class can implement multiple interfaces. For example:

```
interface Interface1 {
    // interface methods
}
interface Interface2 {
    // interface methods
}
class MyClass implements Interface1, Interface2 {
    // class members
}
```

Here, MyClass implements both Interface1 and Interface2.

**C: Python also supports multiple inheritance directly.** A class can inherit from multiple base classes. For example:

```
class Base1:
    pass
class Base2:
    pass
class Derived(Base1, Base2):
    pass
```

Here, Derived inherits from both Base1 and Base2.

**D: VB.NET, similar to Java, does not support multiple inheritance of classes directly.** It supports multiple interface inheritance, where a class can implement multiple interfaces.

**E: The original BASIC language typically does not support object-oriented features like inheritance and multiple inheritance.**

**The Final Answer is: Option B**

**Q24 Text Solution:**

Option A: Accuracy means the possibility for software to do the outputs originally intended.  
 Option B: Specificity measures the ability of a test to detect the absence of the condition when it is not present. It is the probability that the test result is negative when the condition is absent.  
 Option C: Defect Detection Efficiency (DDE) of a phase is the ratio of the number of defects detected in a phase to the total number of defects, expressed as a percentage. DDE is used to measure the effectiveness of each phase.  
 Option D: Throughput refers to the rate at which work items or features are completed and delivered. It measures the productivity of a team or project by quantifying the amount of output produced within a given period.  
 So option C is the correct answer.

**Q25 Text Solution:**

RAID (Redundant Arrays of Independent Disks) is a technique that makes use of a combination of multiple disks for storing the data instead of using a single disk for increased performance, data redundancy, or to protect data in the case of a drive failure.

RAID 2(Error-correction coding): RAID 2 records an Error Correction Code (ECC) using hamming code parity.

RAID 3 (bit interleaved parity organization): It strips the data onto multiple disks. The parity bit is generated for each disk section and stored on a different dedicated disk.

RAID 4: RAID 4 consists of Block-level Striping. At this level, the entire set or block of data is written onto the data disk, and then the parity is generated and stored on a different set of the disk.

RAID-6 (Block-Level Striping with two Parity Bits) Reed Solomon codes operate by dividing a message into blocks, calculating redundant parity data for each block based on the message contents, and transmitting the blocks with parity. So option B is the correct answer.

**Q26 Text Solution:**

Given  $a_n = 8 a_{n-1} - 16 a_{n-2}$



Statement A:  $a_n=1$

putting the value of  $a_n=1$  in eq -(1)

$$1=8.1-16.1 = -8$$

So statement A is the not solution of given recurrence relation.

Statement B:  $a_n=2^n$

putting the value of  $a_n=2^n$  in eq -(1)

$$2^n = 8.2^{n-1} - 16.2^{n-2}$$

$$2^n = 8.2^{n-1} - 8.2^{n-1}$$

$$2^n = 0.$$

So statement B is also not the solution of a given recurrence relation.

Statement C.  $a_n=4^n$

$$4^n = 8.4^{n-1} - 16.4^{n-2}$$

Diving the above equation by  $4^{n-2}$

$$4^2 = 8.4 - 16$$

$$16 = 16$$

So statement C is a solution of a given recurrence relation.

Statement D:  $a_n=n.4^n$

$$n.4^n = 8.(n-1)4^{n-1} - 16.(n-2).4^{n-2}$$

Diving the above equation by  $4^{n-2}$

$$n.4^2 = 8(n-1).4 - 16(n-2)$$

$$16n = 32(n-1) - 16(n-2)$$

$$16n = 32n - 32 - 16n + 32$$

$$16n = 16n$$

So statement D is also a solution of a given recurrence relation.

Statement E:  $a_n=(-4)^n$

$$(-4)^n = 8.(-4)^{n-1} - 16.(-4)^{n-2}$$

$$16 = -32 - 16 = -48$$

So statement E is also not the solution of a given recurrence relation.

Thus option C is the correct answer.

#### Q27 Text Solution:

A. Thread spawn - IV. thread is provided with its own register context.

Explanation: Thread spawn refers to the creation of a new thread. When a thread is spawned, it typically involves allocating resources including its own register context (CPU registers where thread-specific data is stored).

B. Block thread - III. processor turns to the execution of another ready thread.

Explanation: Blocking a thread refers to the situation where a thread is paused or delayed, often due to waiting for some resource or event. When a thread is blocked, the processor can switch to execute another thread that is ready to run.

C. Jacketing - I. to convert a blocking system call into a nonblocking call.

Explanation: Jacketing refers to the technique used to convert a blocking system call (which halts the thread until it completes) into a nonblocking call (where the thread can continue execution without waiting). This typically involves using asynchronous mechanisms.

D. Flush - II. to reclaim pages.

Explanation: Flushing typically refers to the process of freeing up or reclaiming resources, such as releasing memory pages that are no longer needed or ensuring that data is written back to persistent storage.

So option C is the correct answer.

#### Q28 Text Solution:

A. HTML - HTML stands for Hyper Text Markup Language. HTML is the standard markup language for creating Web pages.

B. XML - Extensible Markup Language (XML) is used to define and store data in a shareable manner. XML supports information exchange between computer systems such as websites, databases, and third-party applications.

C. DHTML- DHTML, or Dynamic HTML, is a web-development technique combining HTML, CSS, and JavaScript to create dynamic, interactive web pages and complex web applications.

LML and E. PML are not standard markup languages.

So option D is the correct answer.

#### Q29 Text Solution:

- Asynchronous Transfer Mode(ATM) network is an International Telecommunication Union-Telecommunications Standards Section (ITU-T) efficient call relay system that sends all information, including multiple service kinds such as data, video, and voice, in small fixed-



size packets known as cell. Cells are transmitted asynchronously, and the network is designed to facilitate connections.

- ATM cell consists of a 5-byte header and a 48-byte payload. ATM defines two different cell formats: user-network interface (UNI) and network-network interface (NNI)

UNI- This is used within private networks of ATMs for communication between ATM endpoints and ATM switches. It includes the Generic Flow Control (GFC) field.

NNI Header: is used for communication between ATM switches, and it does not include the Generic Flow Control (GFC) instead it includes a Virtual Path Identifier (VPI) which occupies the first 12 bits.

So option B is the correct answer.

**Q30 Text Solution:**

Given  $A = \{(x_1, 0.3), (x_2, 0.7), (x_3, 0.4)\}$

The complement of A would be  $1 - \mu_A =$

$\{(x_1, 0.7), (x_2, 0.3), (x_3, 0.6)\}$

So option C is the correct answer.

**Q31 Text Solution:**

Option A) These instructions specify three operands or addresses, which may be memory locations or registers. The instruction operates on the contents of all three operands, and the result may be stored in the same or a different location.

Here  $X = (A+B) * (C+D)$

R1, R2 are registers

$M[]$  is any memory location

ADD R1, A, B equivalent to  $R1 = M[A] + M[B]$

ADD R2, C, D equivalent to  $R2 = M[C] + M[D]$

MUL X, R1, R2 equivalent to  $M[X] = R1 * R2$

Option B) 2 - Address instructions - These instructions specify two operands or addresses, which may be memory locations or registers. The instruction operates on the contents of both operands, and the result may be stored in the same or a different location. For example, a two-address instruction might add the contents of two registers together and store the result in one of the registers.

Option C) 1 - Address instructions - These instructions specify one operand or address, which typically refers to a memory location or register. The instruction operates on the contents of that operand, and the result may be stored in the same or a different location. For example, a one-address instruction might load the contents of a memory location into a register.

option D) RISC instructions - A Reduced Instruction Set Computer is a type of microprocessor architecture that utilizes a small, highly-optimized set of instructions rather than the highly-specialized set of instructions found in other architectures. It uses only Hardwired control unit.

So option A is the correct answer.

**Q32 Text Solution:**

A grammar is said to be ambiguous if there exists more than one leftmost derivation or more than one rightmost derivation or more than one parse tree for the given input string. If the grammar is not ambiguous, then it is called unambiguous.

Let us consider a grammar G with the production rule

$E \rightarrow I$

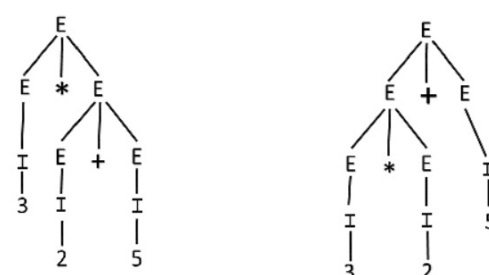
$E \rightarrow E + E$

$E \rightarrow E * E$

$E \rightarrow (E)$

$I \rightarrow \varepsilon \mid 0 \mid 1 \mid 2 \mid \dots \mid 9$

For the string " $3 * 2 + 5$ ", the above grammar can generate two parse trees by leftmost derivation:



Since there are two parse trees for a single string " $3 * 2 + 5$ ", the grammar G is ambiguous.

So option C is the correct answer.

NOTE- Option B can also be the right answer because a grammar is said to be ambiguous if there exists more than one leftmost derivation or



more than one rightmost derivation or more than one parse tree for the given input string.

**Q33 Text Solution:**

Backpropagation is a key technique used in training artificial neural networks (ANNs). It involves adjusting the weights of the network by propagating the error backward from the output layer to the input layer. So it is a learning technique that adjusts weights in the neural network by propagating weight changes backward from sink to source.

So option B is the correct answer.

**Q34 Text Solution:**

$F(A, B, C) = (0, 2, 6)$

$d(A, B, C) = (1, 3, 5)$

K-Map constructed from Given Expression is:

| $a \backslash b, c$ | 00 | 01 | 11 | 10 |
|---------------------|----|----|----|----|
| 0                   | 1  | -  | -  | 1  |
|                     | 0  | 1  | 3  | 2  |
| 1                   | 0  | -  | 0  | 1  |
|                     | 4  | 5  | 7  | 6  |

Let us make the pairs

| $a \backslash b, c$ | 00 | 01 | 11 | 10 |
|---------------------|----|----|----|----|
| 0                   | 1  | -  | -  | 1  |
|                     | 0  | 1  | 3  | 2  |
| 1                   | 0  | -  | 0  | 1  |
|                     | 4  | 5  | 7  | 6  |

from the above K-map we can see one quad (0,1,3,2) and one pair (2,6) is there.

So the Boolean expression will be option D  $A' + BC'$  is the correct answer.

**Q35 Text Solution:**

A. In computer programming, a statement is a syntactic unit of an imperative programming language that expresses some action to be carried out.

B. Once the program is written we will compile the program.

D. Variables used in the program have types, which determine the kind of data that can be

stored in them and the operations that can be performed on them. Type checking and type conversions are important during program execution at compile time.

C. Memory Allocation- During runtime, memory needs to be allocated to store variables, data structures, etc., that the program will use. This is typically managed by the runtime environment.

E.: After the program finishes execution, memory that was allocated dynamically needs to be deallocated to prevent memory leaks and free resources for reuse. Memory deallocation is also managed by the runtime environment.

So option B is the correct answer.

**Q36 Text Solution:**

To calculate the average memory access time given the cache access time, main memory access time, and hit ratio, we use the following formula:

Average Access Time (AAT) = Hit ratio  $\times$  Cache access time + (1 - Hit ratio)  $\times$  Main memory access time

Given:

Cache access time = 100 ns

Main memory access time = 1100 ns

Hit ratio = 0.9

Now, substitute these values into the formula:

Average Access Time (AAT) =  $0.9 \times 100$  ns +  $0.1 \times 1100$  ns

Calculate each term:  $0.9 \times 100$  ns = 90 ns

$0.1 \times 1100$  ns = 110 ns

Now, sum these values to get the average access time:

AAT = 90 ns + 110 ns

AAT = 200 ns

Therefore, the average access time is 200 ns

So option A is the correct answer.

**Q37 Text Solution:**

The hexadecimal digits provided represent the beginning of an IPv4 packet header. Let's break down the information and determine the details:

Given hexadecimal digits:

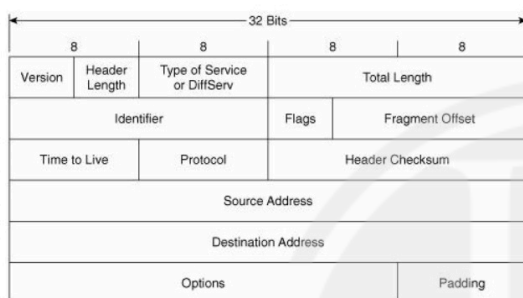
5400001800100110040601...

IPv4 Header Format:



The first 20 bytes (160 bits) of an IPv4 packet header contain various fields, including:

- Version (4 bits)
- Internet Header Length (IHL) (4 bits)
- Type of Service (TOS) or Differentiated Services Code Point (DSCP) (8 bits)
- Total Length (16 bits)
- Identification (16 bits)
- Flags (3 bits)
- Fragment Offset (13 bits)
- Time to Live (TTL) (8 bits)
- Protocol (8 bits)



Extracting Information:

Time To Live (TTL): It is of 8-bit field. This field indicates the maximum time the datagram is allowed to remain in the internet system. From the given hexadecimal digits 5400001800100110040601..., we can identify the fields as follows:

- Version: 5 (since 5 in hex is 0101 in binary, indicating IPv4)
- IHL: 4 (since 4 in hex is 0100 in binary, indicating the header length in 32-bit words)
- TTL: 04 (hexadecimal 04 is in binary, which is 4 in decimal)
- The upper-layer protocol is identified by the Protocol field in the IPv4 header. This field is located after the TTL field and specifies the protocol used in the data portion of the packet (e.g., TCP, UDP, ICMP).

IPv4 Header and Protocol Field:

- The IPv4 header contains a 1-byte (8-bit) Protocol field. This field indicates the protocol used in the data part of the IPv4 packet.

- The Protocol field values are standardized and assigned to various protocols. For example:
- 1 for ICMP (Internet Control Message Protocol)
- 6 for TCP (Transmission Control Protocol)
- 17 for UDP (User Datagram Protocol) and others.

So option D is the correct answer.

### Q38 Text Solution:

To determine the number of spanning trees for each graph and arrange them accordingly:

$K_3$  (Complete Graph with 3 nodes):

$K_3$  has 3 nodes and every pair of distinct nodes is connected by a unique edge.

The number of spanning trees in  $K_3$  is  $3^{3-2} = 3$  because removing any one edge from  $K_3$  results in a spanning tree.

$K_4$  (Complete Graph with 4 nodes):

$K_4$  has 4 nodes, and the number of spanning trees in  $K_4$  is  $4^{4-2} = 16$ . This follows from the Matrix-Tree Theorem or Cayley's formula, which states that the number of spanning trees in a complete graph with  $n$  nodes is  $n^{n-2}$ .

$K_{2,2}$  (Complete bipartite graph with 2 nodes in each partition):

$K_{2,2}$  has 4 nodes (2 nodes in each of 2 partitions), and the number of spanning trees in  $K_{2,2}$  is  $2^{2-1} \times 2^{2-1} = 4$ . This can be calculated by considering spanning trees that cover each partition independently.

$C_5$  (Cycle graph with 5 nodes):

$C_5$  has 5 nodes arranged in a cycle, and the number of spanning trees in  $C_5$  is 5, calculated using the formula for spanning trees in a cycle graph  $n$ .

Now, let's arrange them based on the number of spanning trees:

Therefore, arranging them in increasing order of the number of spanning trees gives us:

- $K_3$  (3 spanning trees)
- $K_{2,2}$  (4 spanning trees)
- $C_5$  (5 spanning trees)
- $K_4$  (16 spanning trees)

So, the correct answer is A, C, D, and B.



**Q39 Text Solution:**

- TDMA is the channelization protocol in which bandwidth of channel is divided into various stations on the time basis.
- HDLC is a bit-oriented, synchronous data link layer protocol created by the International Organization for Standardization (ISO).
- Pure ALOHA is a simple and early random access protocol. In this protocol, any device can transmit data anytime without checking whether the channel is in use.
- There are three types of controlled access protocols. These are reservation, polling, and token passing. All these three types are used in transferring data.

So option B is the correct answer.

**Q40 Text Solution:**

(1) Propagation delay

$d = 5000\text{km}$

$s = 2 \times 10^8 \text{ m/s}$

$T_{\text{prop}} = \text{Distance} / \text{speed} = 5000 \times 10^3 / 2 \times 10^8$   
 $= 0.025 \text{ sec}$

(2) Transmission Time  $T_{\text{trans}} = \text{Frame}$

$\text{size} / \text{Bandwidth}$

$= 10 \times 10^6 / 6 \times 10^6$

$= 10/6 = 1.6667 \text{ sec}$

(3) Queuing delay  $T_{\text{queue}} = 20 \times 2 = 40 \mu\text{s} =$   
 $0.04\text{ms}$

(4) Processing delay  $T_{\text{proc}} = 20 \times 1 = 20 \mu\text{s} = 0.02\text{ms}$

(5) Total delay  $= T_{\text{prop}} + T_{\text{trans}} + T_{\text{queue}} + T_{\text{proc}}$   
 $= 0.025\text{sec} + 1.6667\text{sec} + 0.04\text{ms} + 0.02\text{ms}$   
 $= 1.69176 \text{ sec}.$

Thus option B is the correct answer.

**Q41 Text Solution:**

To determine the total time required for an algorithm performing  $2^{50}$  bit operations, where each bit operation takes  $2^{-38}$  seconds, follow these steps:

Calculate the total time:  $T$  can be calculated as:

$T = (2^{50}) \times 2^{-38} \text{ seconds}$

Simplify the exponentiation:  $= 2^{50-38} = 2^{12} =$   
 $4096 \text{ seconds}$

Therefore, the total time  $= 4096/60 \times 60 = 1.1389$  hours

Which nearly equal to 1 hour, so option A is the correct answer.

**Q42 Text Solution:**

Duality theorem states that the dual of the Boolean function is obtained by interchanging the logical AND operator with logical OR operator and zeros with ones. Correct match for the given List is as follows.

|                                      |                                             |
|--------------------------------------|---------------------------------------------|
| A. $x.y+0$                           | I. $x+y.1$                                  |
| B. $\bar{x} \cdot 1 + (\bar{y} + z)$ | II. $(\bar{x} + 0) \cdot (\bar{y} \cdot z)$ |
| C. $\bar{x} \cdot (\bar{y} + 0)$     | III. $\bar{x} + \bar{y} \cdot 1$            |
| D. $x \cdot 1 + (y + \bar{z})$       | IV. $(x + 0) \cdot (y \cdot \bar{z})$       |
|                                      |                                             |

So option C is the correct answer.

**Q43 Text Solution:**

Given A bit string of length four is generated at random. It contains at least two consecutive 0's given that the first bit is 0.

These strings can be represented as 0abc, where a,b,c are binary digits 0 or 1.

Total number of such strings is 0abc so there are  $2 \times 2 \times 2 = 8$  such strings.

Now, we need to find the number of strings that contain at least two consecutive 0s. These strings are:

0000

0001

0010

0011

0100

So above 5 strings meet this criterion.

Now Probability = Number of favorable outcomes / Total number of outcomes  
 $= \frac{5}{8}$

Thus option C is the correct answer.

**Q44 Text Solution:**

In computer networking, Point-to-Point Protocol (PPP) is a data link layer (layer 2) communication protocol that connects two routers directly, with no host or other networking in between.



The main services provided by Point-to-Point Protocol are –

- Defining the frame format of the data to be transmitted.
- Defining the procedure of establishing a link between two points and exchange of data.
- Stating the method of encapsulation of network layer data in the frame.
- Stating authentication rules of the communicating devices.
- Providing an address for network communication.
- Providing connections over multiple links.
- Supporting a variety of network layer protocols by providing a range of services.

So option A is the correct answer.

**Q45 Text Solution:**

Tree Based Protocol are:

- Partial Order on Database items determines a tree-like structure.
- Only Exclusive Locks are allowed.
- The first lock by  $T_i$  may be on any data item. Subsequently, a data  $Q$  can be locked by  $T_i$  only if the parent of  $Q$  is currently locked by  $T_i$ .
- Data items can be unlocked at any time. Tree based Protocol ensures Conflict Serializability and Deadlock Free schedule. We need not wait to unlock a Data item thus increasing the concurrency.
- shortened wait times and increased concurrency.
- The protocol is deadlock-free; thus, no rollbacks are necessary.

So option A is the correct answer.

**Q46 Text Solution:**

The cardinality of a set is defined as the number of elements in a mathematical set.

$A_1 = \{\{1, 2\}, \{3\}\}$ , Here in set 2 elements are there so cardinality of  $A_1 = 2$

$A_2 = \{\{1\}, \{2\}, \{3\}, \{4\}\}$ , Here in set 4 elements are there so cardinality of  $A_2 = 4$

$A_3 = \{\{1, 2, 3, 4, 5, 6\}\}$ , Here in set 1 element is there so cardinality of  $A_3 = 1$

$A_4 = \{\{1, 2\}, \{2, 3, 4\}, \{5\}\}$ , Here in set 3 elements are there so cardinality of  $A_4 = 3$

$A_5 = \{\{1\}, \{2\}, \{3\}, \{4\}, \{5\}\}$ , Here in set 5 elements are there so cardinality of  $A_5 = 5$

So option D is the correct answer.

**Q47 Text Solution:**

Cardinality of a fuzzy set is the sum of the membership values of all elements of the fuzzy set. The membership value of an element in that fuzzy set can vary from 0.0 to 1.0. So sum of the membership values of all elements of the fuzzy set is finite. Thus option B is the correct answer.

**Q48 Text Solution:**

Reverse Engineering Goals:

- Cope with Complexity: Reverse engineering is a common tool used to understand and control system complexity. It gives engineers the ability to analyze complex systems and reveal details about their architecture, relationships and design patterns.
- Recover lost information: Reverse engineering seeks to retrieve as much information as possible in situations where source code or documentation are lost or unavailable. Rebuilding source code, analyzing data structures and retrieving design details are a few examples of this.
- Detect side effects: Understanding a system or component's behavior requires analyzing its side effects. Unintended implications, dependencies, and interactions that might not be obvious from the system's documentation or original source code can be found with the use of reverse engineering.
- Synthesis higher abstraction: Abstracting low-level features in order to build higher-level representations is a common practice in reverse engineering. This abstraction makes communication and analysis easier by facilitating a greater understanding of the system's functionality.



- Facilitate Reuse: Reverse engineering can be used to find reusable parts or modules in systems that already exist. By understanding the functionality and architecture of a system, developers can extract and repurpose components for use in other projects, improving efficiency and decreasing development time.

Data flow is a testing method that is used to find the test paths of a program according to the locations of definitions and uses of variables in the program.

So option D is the correct answer.

#### Q49 Text Solution:

To determine the maximum possible chairs that can be labeled in the auditorium starting from A01 and following the format of a letter followed by a positive two-digit integer, we need to consider the range of possible labels.

The format of the label is:

A letter (A-Z)

Two digits (01-99)

Let's calculate the total number of possible labels:

Letters (A-Z):

There are 26 letters in the English alphabet (A, B, C, ..., Z).

Two-digit numbers (01-99):

There are 99 possible two-digit numbers starting from 01 to 99.

Now, multiply the number of options for letters by the number of options for two-digit numbers:  $26 \times 99 = 2574$

Therefore, the maximum number of possible chairs that can be labeled in this way, starting from A01 and following the given format, is option B, 2574.

#### Q50 Text Solution:

The correct order for constructing a finite automaton progresses as follows:

Problem Statement (D): Understand the problem that the automaton needs to solve, including the language it should accept or recognize.

Regular Expression (B): Express the language that the automaton should recognize using a regular expression. Regular expressions are a concise way to describe patterns in strings.

NFA- $\epsilon$  (C): Construct a nondeterministic finite automaton (NFA) based on the regular expression. NFAs allow multiple transitions from a state on the same input symbol and can have  $\epsilon$ -transitions (epsilon transitions).

NFA- to DFA Conversion: Convert the NFA to a deterministic finite automaton (DFA). This conversion eliminates  $\epsilon$ -transitions and nondeterminism, resulting in a DFA.

Minimum State DFA (A): Minimize the DFA obtained from the previous step. Minimization reduces the number of states while preserving the language recognized by the automaton. So option B is the correct answer.

#### Q51 Text Solution:

Option A) In which all the non-zero elements lie only on the leading diagonal.

This describes a diagonal matrix, where all non-zero elements are on the main diagonal.

Option B) In which all the non-zero elements lie above leading diagonal.

This describes an upper-triangular matrix, where all non-zero elements are above the main diagonal.

Option C) In which all the non-zero elements lie below leading diagonal.

This describes a lower-triangular matrix, where all non-zero elements are below the main diagonal.

Option D) Which is not defined for sparse matrix.

This is not correct, as sparse matrices can be defined and manipulated regardless of their structure (whether they are lower-triangular, upper-triangular, diagonal, etc.).

So option C is the correct answer.

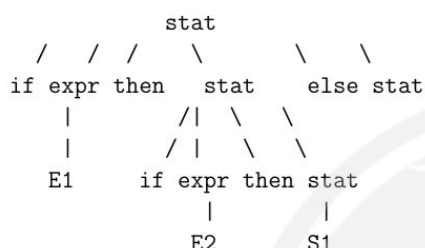
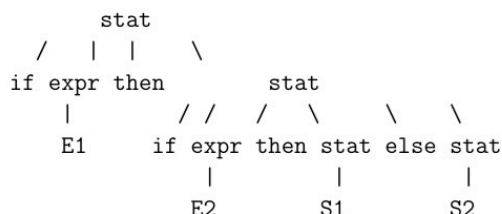
#### Q52 Text Solution:

A grammar that produces more than one parse tree for some sentence is ambiguous. Or:

- If a grammar has more than one leftmost derivation for a single sentential form, the grammar is ambiguous.



- If a grammar has more than one rightmost derivation for a single sentential form, the grammar is ambiguous.  
The sentence if E1 then if E2 then S1 else S2 has the following two different parse trees:



**Left factoring:** Left factoring is a technique used to eliminate common prefixes in productions to resolve potential conflicts in parsing. This grammar does not exhibit left factoring issues because each production is distinct and does not share common prefixes that would cause a parsing conflict.

**Left recursion:** Left recursion occurs when a non-terminal A directly or indirectly derives a string that starts with itself, i.e.,

$A \Rightarrow * A\alpha$  for some string  $\alpha$ . The given grammar does not have left recursion because the leftmost symbol in each production is a terminal symbol ("if") or distinct non-terminals.

**$\lambda$ -moves (epsilon moves):**  $\lambda$ -moves are generally associated with regular expressions and finite automata rather than context-free grammars. This option is not applicable to the context of context-free grammars and parsing ambiguities. So option A is the correct answer.

### Q53 Text Solution:

Virtual Memory Segmentation in OS is a memory management technique used in computer systems. It divides processes into multiple

segments, with the number of segments represented by 'n'.

Virtual Memory Segmentation refers to the process of loading a process by loading all of its segments into dynamic partitions that need not be contiguous. Virtual memory segmentation is the partitioning of memory into various segments. So option A is correct.

Virtual memory allows a single process to have a virtual address space that is not continuous and, in reality, is not entirely contained in physical memory. The segments of virtual memory correspond to different parts of the process.

There is no internal fragmentation as the segments are allocated the space as much as it require, provided the memory is big enough to accommodate that segment. Unlike fixed partitioning, memory is not wasted in segmentation. So option B is correct.

Virtual address space refers to the room of addresses which usually refers to the reference to the different slots of virtual memory allotted to different processes. Operating systems allocate this set of addresses for the processes to use in order to access their designated virtual memory. Virtual address space enables dynamic memory allocation as it is mainly needed to carry out the task of assigning memory blocks to the processes when they request dynamically. So it has a large virtual address space.

Virtual address space avoids interference between processes' memory spaces which helps to increase the stability of the system. It helps to make efficient usage of the memory as it helps to share the same memory space for two/more processes. so it can provide a higher degree of multiprogramming.

Thus option D is the correct answer.

### Q54 Text Solution:

A. Union is an user defined datatype in C programming language. It is a collection of variables of different datatypes in the same memory location.



B. Subroutine/Function is a programming construct that allows a programmer to associate a given set of instructions with a specific name.

C. Virtual Reality (VR) provides an interactive environment that simulates a real or imaginary environment, allowing users to interact with virtual objects and experience immersive environments.

D. A shadow mask is used in CRT (Cathode Ray Tube) displays which are output devices to direct electron beams to specific points on the screen. It helps in creating images by controlling where the electron beams hit the phosphor coating inside the CRT.

So option A is the correct answer.

### Q55 Text Solution:

To determine which operation forms a group on the set of integers  $Z$ , we need to check the group axioms:

Closure: For all  $a, b \in Z$ , the operation  $a * b$  must result in another integer.

Associativity: The operation must be associative:

$$(a * b) * c = a * (b * c) \text{ for all } a, b, c \in Z.$$

Identity Element: There exists an identity element  $e \in Z$  such that  $a * e = e * a = a$  for all  $a \in Z$ .

Inverse Element: For each  $a \in Z$ , there exists an inverse element  $a^{-1} \in Z$  such that  $a * a^{-1} = a^{-1} * a = e$ .

Let's analyze each operation given:

$$(1) *(a, b) = a + b + 2$$

Closure:  $a + b + 2$  is an integer for all  $a, b \in Z$ .

Associativity:  $(a + b + 2) + c + 2 = a + (b + c + 2) + 2$ , which is associative.

Identity Element:  $e = -2$  because  $a + (-2) + 2 = a$  for all  $a \in Z$ .

Inverse Element:  $a^{-1} = -a - 4$  because  $a + (-a - 4) + 2 = -2$ , the identity element.

Therefore,

$\langle Z, * \rangle$  forms a group with operation  $*(a, b) = a + b + 2$ .

$$(2) *(a, b) = 2a + b$$

Closure:  $2a + b$  is an integer for all  $a, b \in Z$ .

Associativity:  $(2a + b) + c = 2a + (b + c)$ , which is associative.

Identity Element: There is no integer  $e \in Z$  such that

$$2a + e = a \text{ for all } a \in Z.$$

Inverse Element: There is no integer  $b \in Z$  such that  $2a + b = 0$  for all  $a \in Z$ .

Thus,  $*(a, b) = 2a + b$  does not form a group because it lacks an identity element and inverses for all elements.

$$(3) *(a, b) = (a + b)/2$$

Closure:  $(a + b)/2$  is not necessarily an integer for all  $a, b \in Z$ .

Associativity:  $(a + b)/2 + c/2$  does not necessarily equal  $a/2 + (b + c)/2$  so it is not associative.

Identity Element: There is no integer  $e \in Z$  such that  $(a + e)/2 = a$  for all  $a \in Z$ .

Inverse Element: There is no integer  $b \in Z$  such that  $(a + b)/2 = 0$  for all  $a \in Z$ .

Therefore,  $*(a, b) = (a + b)/2$  does not form a group due to issues with closure and associativity.

$$(4) *(a, b) = a^2 + b$$

Closure:  $a^2 + b$  is an integer for all  $a, b \in Z$ .

Associativity:  $(a^2 + b) + c = a^2 + (b + c)$ , which is associative.

Identity Element: There is no integer  $e \in Z$  such that  $a^2 + e = a$  for all  $a \in Z$ .

Inverse Element: There is no integer  $b \in Z$  such that  $a^2 + b = 0$  for all  $a \in Z$ .

Therefore,  $*(a, b) = a^2 + b$  does not form a group because it lacks an identity element and inverses for all elements.

So option A is the correct answer.

### Q56 Text Solution:

A user can assign privileges to any user on any relation" is generally false because:

- Users typically cannot assign privileges to any relation (table). The ability to grant privileges is usually restricted to objects that they own or have been explicitly granted permission to manage.
- Only users with administrative rights (like DBAs) can generally assign privileges to any database object.



A user can assign privileges to any user on its own relations only is True because:

- Users without administrative rights are restricted to granting permissions only on objects they own or have been explicitly granted control over.
- This restriction may vary depending on the database system and the permissions model implemented

The system error that leads to accessibility to the portion of the database is an example of a malicious attack.

- This statement is false. A system error or a bug that inadvertently exposes part of the database is typically considered a non-malicious event or a security incident. Malicious attacks involve intentional actions by unauthorized individuals or entities to compromise the system's security.

Passwords itself need to be secured from unauthorized access.

- This statement is true. Passwords are critical for authentication and access control. They should be stored securely.

So option A is the correct answer.

#### Q57 Text Solution:

Option A) Real-time scheduling policy: Specialized embedded operating systems often prioritize real-time scheduling policies. This means they ensure that tasks are completed within specific time constraints, making them suitable for applications where timing predictability is crucial, such as in industrial control systems or automotive electronics.

Option B) Responds to external interrupts: Embedded operating systems need to respond to external events or interrupts from various hardware components or sensors. This responsiveness is essential for real-time processing and event-driven applications.

Option C) Provides special non-sequential files:

Embedded operating systems typically do not focus on providing special types of files like non-sequential files. The emphasis is more on real-time performance, efficient resource utilization, and responsiveness to external events. File management in embedded systems tends to be simpler and more focused on direct access to data rather than complex file structures.

Option D) Provides fixed or variable-sized partitions:

Embedded systems often manage memory resources through fixed or variable-sized partitions. This allows efficient utilization of limited resources and helps in optimizing memory allocation for specific tasks or applications running on the embedded system.

So option C is the correct answer.

#### Q58 Text Solution:

Frame relay is a packet mode communication service that is designed to manage enhanced WANs.

Each Frame Relay protocol data unit (PDU) consists of the following fields in the following order:

A. Flag Field. The flag is used to perform high-level data link synchronization which indicates the beginning and end of the frame with the unique pattern 01111110.

C. Address Field. Each address field may occupy either octet 2 to 3, octet 2 to 4, or octet 2 to 5, depending on the range of the address in use.

D. Information Field. A system parameter defines the maximum number of data bytes that a host can pack into a frame. Hosts may negotiate the actual maximum frame length at call set-up time.

B. Frame Check Sequence (FCS) Field. Since one cannot completely ignore the bit error-rate of the medium, each switching node needs to implement error detection to avoid wasting bandwidth due to the transmission of erred frames.

Thus option A is the correct answer.

#### Q59 Text Solution:



Option A) The basic timestamp ordering protocol (BASIC T/O) allows reads and writes on database objects without using locks. Instead, every database object X is tagged with a timestamp of the last transaction that successfully performed a read (denoted as R-TS(X)) or write (denoted as W-TS(X)) on that object. The DBMS then checks these timestamps for every operation. If a transaction tries to access an object in a way that violates the timestamp order, the transaction is aborted and restarted.

For dynamic databases in which transactions perform insertions, updates, and deletions time stamping will not be sufficient.

The phantom phenomenon arises when two transactions logically conflict even though they do not lock a common data item. In a database, the phantom problem is a scenario where a transaction reads a set of rows from the database that match a certain condition, and when the same transaction attempts to read the rows again (using the same condition), it discovers additional rows that weren't there previously. These new rows are known as 'phantoms'.

Approaches to Address the Phantom Problem:

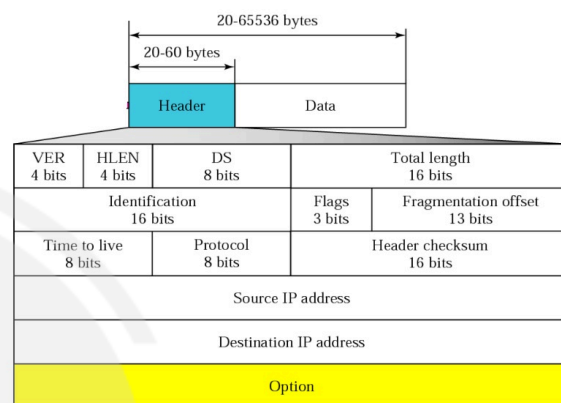
1. Re-Execute Scans: Transactions may re-run queries at commit time to check for different results, indicating missed changes due to new or deleted records.
2. Predicate Locking: This involves acquiring locks based on the predicates of the queries, ensuring that any data that satisfies the predicate cannot be modified by other transactions. A predicate lock in database systems is a type of lock that prevents the modification of all data rows that satisfy a specific condition, thus ensuring data integrity when multiple transactions access the same data simultaneously.
3. Index Locking: Utilizing index keys to protect ranges of data, preventing phantoms by ensuring that no new data can fall within the locked ranges. After an index range is locked, no other transaction can insert, update, or delete any row within this range until the transaction that holds

the lock completes. In this way, index locking ensures the consistent state of a transaction, resolving the phantom problem.

Option D) Indexing often refers to the process of creating and maintaining data that allow for efficient access to elements based on some key or index. For dynamic database concurrency problems like phantom problem indexing is not alone sufficient.

So option B is the correct answer.

#### Q60 Text Solution:



#### IPv4 Datagram Header

- **VERSION:** Version of the IP protocol (4 bits), which is 4 for IPv4.
- **HLEN:** IP header length (4 bits), which is the number of 32 bit words in the header. The minimum value for this field is 5 and the maximum is 15.
- **Type of service:** Low Delay, High Throughput, Reliability (8 bits)
- **Total Length:** Length of header + Data (16 bits), which has a minimum value 20 bytes and the maximum is 65,535 bytes.
- **Identification:** Unique Packet Id for identifying the group of fragments of a single IP datagram (16 bits)
- **Flags:** 3 flags of 1 bit each : reserved bit (must be zero), do not fragment flag, more fragments flag (same order)
- **Fragment Offset:** Represents the number of Data Bytes ahead of the particular fragment in the particular Datagram. Specified in terms of



number of 8 bytes, which has the maximum value of 65,528 bytes.

- Time to live: Datagram's lifetime (8 bits), It prevents the datagram to loop through the network by restricting the number of Hops taken by a Packet before delivering to the Destination.
- Protocol: Name of the protocol to which the data is to be passed (8 bits)
- Header Checksum: 16 bits header checksum for checking errors in the datagram header
- Source IP address: 32 bits IP address of the sender
- Destination IP address: 32 bits IP address of the receiver
- Option: Optional information such as source route, record route. Used by the Network administrator to check whether a path is working or not.

So option A is the correct answer.

#### Q61 Text Solution:

Cyclomatic complexity =  $E - N + 2 \cdot P$

where,

E = number of edges in the flow graph.

N = number of nodes in the flow graph.

P = number of nodes that have exit points

Here E= 11, N=9, P=1

Cyclomatic complexity =  $11 - 9 + 2 = 4$

So option B is the correct answer.

#### Q62 Text Solution:

IP (Internet Protocol) is responsible for host-to-host communication, while TCP (Transmission Control Protocol) is responsible for process-to-process communication.

Here's a breakdown of what each term means in this context:

IP (Internet Protocol):

- IP is a fundamental protocol in the Internet protocol suite (TCP/IP).
- It provides the addressing and routing mechanism that enables packets to be routed across networks.

- IP handles the addressing and routing of packets between hosts (computers or devices) on a network.
- Its primary responsibility is to deliver packets from the source host to the destination host based on IP addresses.

TCP (Transmission Control Protocol):

TCP is a protocol that operates at a higher level than IP in the TCP/IP stack.

- It provides reliable, ordered, and error-checked delivery of a stream of bytes between applications running on hosts communicating over an IP network.
- It manages process-to-process communication by establishing a connection between the sender and receiver, managing data flow, handling acknowledgments, and retransmitting lost packets if necessary.

The data link layer is responsible for the node-to-node delivery of the message. The main function of this layer is to make sure data transfer is error-free from one node to another, over the physical layer. The socket is primarily a concept used in the transport layer of the Internet protocol suite or session layer of the OSI model.

So option D is the correct answer.

#### Q63 Text Solution:

Formulation Mathematically (B):

This is the initial step where the problem is translated into mathematical terms.

Mathematical equations are formulated to represent the problem, including variables, constraints, and the objective function.

Solve Mathematically (C):

Once the mathematical model is formulated, the next step is to solve it using appropriate mathematical techniques. This could involve optimization algorithms, numerical methods, or simulations depending on the nature of the model and problem.

Solution fitting with data (A):

After obtaining mathematical solutions, the model needs to be compared and adjusted to fit



real-world data. This step involves calibration or validation of the model against empirical data to ensure its accuracy and relevance.

Repeat formulation if it fits worst with data (D): If the initial model does not fit well with the data or if discrepancies are found during the fitting process, it may be necessary to revise the formulation. This step involves refining the model parameters, assumptions, or structure to improve its alignment with observed data.

Interpretation results (E):

Finally, after the model is successfully calibrated and validated, the results are interpreted in the context of the original problem. This step involves analyzing the implications of the model's findings, drawing conclusions, and making recommendations based on the results obtained. So option A is the correct answer.

#### Q64 Text Solution:

Windows uses a priority-based scheduling algorithm to manage processes running on the system. This prioritization helps determine which processes get access to CPU time and in what order. Windows categorizes priorities into two main types: real-time priorities and non-real-time priorities.

- Real-time priorities are used for critical system tasks that require immediate attention and are typically reserved for system-level processes. They have a higher priority than non-real-time priorities.
- Non-real-time priorities are used for user-level processes and applications. These priorities are further subdivided into 16 levels.

Here's how the 16 non-real-time priority levels are structured:

- They range from 0 (lowest) to 15 (highest).
- Each level represents a different priority at which a process can run relative to other non-real-time processes.
- Higher priority levels (closer to 15) indicate more urgency and readiness to execute,

whereas lower levels (closer to 0) indicate less urgency and lower priority.

- These priorities are dynamic and can change based on various factors such as process behavior, resource availability, and system load.
- The scheduling algorithm in Windows adjusts these priorities to ensure fair access to CPU resources while prioritizing processes that need more immediate attention.

So option D is the correct answer.

#### Q65 Text Solution:

The I/O methods in which memory addresses and I/O addresses are distinct and referred to as Isolated I/O. This method uses separate address spaces for memory and I/O operations, ensuring that memory operations do not interfere with I/O operations and vice versa.

So, the correct answer based on the options provided is Statement A, B Only:

A. Isolated I/O

B. Memory-Mapped I/O

Isolated I/O (Option A): This directly addresses the question as it describes the method where memory addresses and I/O addresses are distinct.

Memory-Mapped I/O (Option B): This method maps I/O devices to the same address space as memory, which contrasts with isolated I/O where separate address spaces are used.

Strobe Control (Option C): This refers to a signal used in certain I/O operations but does not specifically relate to the addressing scheme.

Handshaking (Option D): This refers to a method of communication between devices to establish readiness or acknowledgment but is not directly related to memory and I/O address separation.

Interrupt (Option E): This refers to a mechanism where a device can signal the CPU to interrupt its current activities to handle a specific event. It is not directly related to the distinction between memory and I/O addresses.

So option A is the correct answer.

#### Q66 Text Solution:



Scanning, parsing, and validating (C):

This is the initial step where the query is scanned to identify its components (tokens), parsed to create a parse tree or syntax tree, and validated to ensure it conforms to the syntax and semantics of the database query language.

Query optimizer (A):

After the query has been parsed and validated, the optimizer analyzes it to generate an efficient execution plan. The optimizer considers various factors such as indexes, join methods, and access paths to determine the best way to execute the query.

Query code generator (D):

Once the query has been optimized, the query code generator transforms the optimized query plan into executable code that the database system can understand and execute.

Runtime database processor (B):

Finally, the runtime database processor executes the generated query code. This involves accessing data from disk or memory, performing operations specified in the query plan (such as joins, aggregations, and sorting), and returning the results to the user or application.

So option B is the correct answer.

#### Q67 Text Solution:

$$(p \vee r) \rightarrow p$$

| p | r | pVr | ~(pVr) | ~(pVr) V p |
|---|---|-----|--------|------------|
| 0 | 0 | 0   | 1      | 1          |
| 0 | 1 | 1   | 0      | 0          |
| 1 | 0 | 1   | 0      | 1          |
| 1 | 1 | 1   | 0      | 1          |

Since all the values of  $(p \vee r) \rightarrow p$  are not 1.

$(p \vee r) \rightarrow p$  is not a tautology.

$$p \rightarrow (p \wedge r)$$

| p | r | p^r | ~p | ~p v (p^r) |
|---|---|-----|----|------------|
| 0 | 0 | 0   | 1  | 1          |
| 0 | 1 | 0   | 1  | 1          |
| 1 | 0 | 0   | 0  | 0          |
| 1 | 1 | 1   | 0  | 1          |

Since all the values of  $p \rightarrow (p \wedge r)$  are not 1.

$p \rightarrow (p \wedge r)$  is not a tautology,

$$\sim p \rightarrow (p \rightarrow r)$$

| p | r | ~p | ~p v r | ~(~p) v (p → r) |
|---|---|----|--------|-----------------|
| 0 | 0 | 1  | 1      | 1               |
| 0 | 1 | 1  | 1      | 1               |
| 1 | 0 | 0  | 0      | 1               |
| 1 | 1 | 0  | 1      | 1               |

Since all the values of  $\sim p \rightarrow (p \rightarrow r)$  are 1.

$\sim p \rightarrow (p \rightarrow r)$  is a tautology.

$$(p \wedge r) \rightarrow (p \rightarrow r)$$

| p | r | (p^r) | ~p v r | ~(p^r) v (p → r) |
|---|---|-------|--------|------------------|
| 0 | 0 | 0     | 1      | 1                |
| 0 | 1 | 0     | 1      | 1                |
| 1 | 0 | 0     | 0      | 1                |
| 1 | 1 | 1     | 1      | 1                |

Since all the values of  $(p \wedge r) \rightarrow (p \rightarrow r)$  are 1.

$(p \wedge r) \rightarrow (p \rightarrow r)$  is a tautology.

$$\sim (p \rightarrow r) \rightarrow p$$

| p | r | ~p v r | ~(~p v r) | ~(p → r) → p |
|---|---|--------|-----------|--------------|
| 0 | 0 | 1      | 0         | 1            |
| 0 | 1 | 1      | 0         | 1            |
| 1 | 0 | 0      | 1         | 1            |
| 1 | 1 | 1      | 0         | 1            |

Since all the values of  $\sim (p \rightarrow r) \rightarrow p$  are 1.

$\sim (p \rightarrow r) \rightarrow p$  is a tautology.

So option D is the correct answer.

#### Q68 Text Solution:

C. CPU Registers - Registers are the fastest because they are directly within the CPU and provide the fastest access to data and instructions.

B. SRAM - SRAM is faster than DRAM because it is static and does not need to be refreshed like DRAM. It is commonly used in cache memory.

A. DRAM - DRAM is slower than SRAM but is still faster than persistent storage devices like hard disks and magnetic tapes. It is used as the main memory in computers.

D. Magnetic tapes - Magnetic tapes are slower compared to RAM and CPU registers because they are sequential access devices.



E. Magnetic tapes are the slowest among the options listed because they rely on mechanical movement to read and write data.

So option B is the correct answer.

**Q69 Text Solution:**

To determine the language accepted by the given Pushdown Automaton (PDA), we need to analyze its transitions and the behavior of its stack.

The PDA transitions are given as follows:

$$\delta(q_0, a, z_0) = (q_0, aaa z_0)$$

$$\delta(q_0, a, a) = (q_0, aaaa)$$

$$\delta(q_0, b, a) = (q_1, \lambda)$$

$$\delta(q_1, b, a) = (q_1, \lambda)$$

$$\delta(q_1, \epsilon, z_0) = (q_f, z_0)$$

Let's interpret these transitions:

- From  $q_0$ , upon seeing an  $a$  with  $z_0$  on the stack, it pushes three  $a$ 's onto the stack.
- From  $q_0$ , upon seeing an  $a$  with  $a$  on the stack, it pushes three  $a$ 's onto the stack.
- From  $q_0$ , upon seeing a  $b$  with  $a$  on the stack, it removes  $a$  from the stack (using  $\lambda$ ).
- From  $q_1$ , upon seeing a  $b$  with  $a$  on the stack, it continues to remove  $a$  from the stack.
- When  $q_1$  reaches  $\epsilon$  (empty input) with  $z_0$  on the stack, it transitions to  $q_f$  and empties the stack.

Now, let's analyze the language recognized by this PDA:

- The PDA starts in state  $q_0$  with  $z_0$  on the stack.
- For every  $a$  read in state  $q_0$ , it adds  $aaa$  to the stack, depending on whether  $z_0$  or  $a$  is on the stack initially.
- For every  $b$  encountered, it removes an  $a$  from the stack.
- When no input is left  $\epsilon$ , if  $z_0$  is on the stack, it accepts (moves to  $q_f$ ).

The key observation here is the balanced nature of the stack operations:

- For each  $a$ , the stack grows by three  $a$ 's.
- For each  $b$ , the stack decreases by one  $a$ .

The language accepted by PDA is

$$\{a^n b^{3n} \mid n > 0\}$$

So option C is the correct answer.

**Q70 Text Solution:**

C. Save the contents of processor registers:

Before executing any critical operation like handling interrupts or subroutines, it's essential to save the current state of the processor's registers. This ensures that the processor's current state is preserved and can be restored later.

D. Check which flag is set: This step involves checking which interrupt flag is set or which subroutine needs to be executed. This check determines the specific task that requires attention.

B. Service the device whose flag is set: Once it's determined which device or subroutine needs attention (based on the flag checked in step D), the corresponding device is serviced or the subroutine is executed.

E. Restore the contents of processor registers: After completing the interrupt handling or subroutine execution, the saved processor register contents (from step C) are restored. This step ensures that the processor returns to its original state before the interrupt or subroutine was handled.

A. Turn on the interrupt facility: This step involves enabling the interrupt mechanism again after the task (interrupt handling or subroutine execution) has been completed. This prepares the system to handle future interrupts.

So option C is the correct answer.

**Q71 Text Solution:**

A. Verification is a process of testing and inspecting products during and after production to collect objective evidence that each one meets all regulatory requirements, fulfills all predetermined specifications, and is ready for use. so it matches with quality assurance.

B. Validation is an essential aspect of quality control that ensures that a product or service meets the customer's requirements and expectations.

C. White box testing involves closely examining the internal logic and structure of code. It focuses on testing individual program paths by



exercising conditions, loops, and logical decisions.

D. Black Box testing technique focuses on testing the functionality of the software without knowing its internal structure. Test cases are derived from the software's requirements and functionality.

So option A is the correct answer.

**Q72 Text Solution:**

- Crossover in GA involves taking two parent solutions and combining parts of them to create new offspring solutions.
- Mutation in GA involves making small random changes to individuals (or solutions) to introduce diversity into the population.
- Selecting a population refers to the process of choosing which individuals will continue to the next generation based on their fitness or other selection criteria. The "Actual count" could play a role in determining how many individuals are selected from the current population to form the next generation.
- Encoding in GA refers to the representation of individuals (solutions) in the genetic algorithm, typically as strings of bits (binary encoding) or other forms suitable for genetic operations like crossover and mutation.

So option C is the correct answer.

NOTE- Multiple options can be correct.

**Q73 Text Solution:**

Option A) In DFA, there is only one path for specific input from the current state to the next state. DFA does not accept the null move, i.e., the DFA cannot change state without any input character. DFA can contain multiple final states. It is used in Lexical Analysis in Compiler.

Machine 1 does not contain multiple final states.

Option B) The finite automata are called NFA when there exist many paths for specific input from the current state to the next state. Every NFA is not DFA, but each NFA can be translated into DFA. NFA is defined in the same way as DFA

but with the following two exceptions, it contains multiple next states, and it contains  $\epsilon$  transition.

So Both are non-equivalent machines. Machine 1 is NFA and Machine 2 is DFA.

Option B is the correct answer.

**Q74 Text Solution:**

The line given is: `for i in range(begin; end; step):`  
To determine the number of tokens in this line, let's break down the components typically recognized in programming languages:

- `for`: This is a keyword indicating the start of a loop.
  - `i`: This is an identifier, representing the loop variable.
  - `in`: This is a keyword indicating the beginning of the iterable range.
  - `range`: This is a built-in function in Python.
  - `(`: This is an opening parenthesis, used to define function arguments or tuples.
  - `begin`: This is an identifier, representing the starting value of the range.
  - `;`: This is a semicolon
  - `end`: This is an identifier, presumably representing the ending value of the range.
  - `;`: This is a semicolon
  - `step`: This is an identifier, presumably representing the step or increment value for the range.
  - `)`: This is a closing parenthesis, closing the function arguments or tuple.
  - `:`: This is colon
- So, the total number of tokens in the line is 12.

**Q75 Text Solution:**

Option A. Conceptual dependency: Conceptual dependency is a semantic representation formalism that focuses on representing the meaning of natural language sentences in terms of conceptual structures. It emphasizes the relationships between actions and objects and is more concerned with semantic content rather than just syntax.

Option B Predicate Logic: Syntactic Nature: Predicate logic, also known as first-order logic, is



highly syntactic. It deals with formal rules governing the syntax and semantics of logical statements. It uses symbols (predicates, variables, quantifiers, and logical connectives) to represent relationships and assertions.

Focus on Syntax: Predicate logic focuses on the formal structure of statements, such as how predicates are applied to arguments, how variables are quantified, and how logical connectives are used to form complex expressions.

Example: In predicate logic, statements like "All men are mortal" can be expressed as  $\forall x (\text{Man}(x) \rightarrow \text{Mortal}(x))$ , where  $\forall$  denotes universal quantification,  $\text{Man}(x)$  and  $\text{Mortal}(x)$  are predicates, and  $\rightarrow$  denotes implication.

Option C Slot-and-Filler:

Syntactic Structure: Slot-and-filler structures, commonly used in frame-based systems, organize knowledge using templates where entities (objects or concepts) are described by filling predefined slots (attributes or properties) with specific values.

Template Structure: Each template defines a syntactic structure consisting of slots (which represent attributes or roles) and fillers (which represent specific values or entities).

Example: In a slot-and-filler representation for a car, a template might have slots like "Manufacturer", "Model", "Color", etc., where each slot is filled with specific information (e.g., "Manufacturer: Toyota", "Model: Camry", "Color: Blue").

Option D: Nonmonotonic systems:

Nonmonotonic logic systems extend classical logic to handle reasoning under uncertainty and with incomplete information.

So option B is the correct answer.

#### Q76 Text Solution:

Option A) Single Point Crossover: A crossover point on the parent organism string is selected. All data beyond that point in the organism string is swapped between the two parent organisms.

Option B) Two-Point Crossover : This is a specific case of a N-point Crossover technique. Two random points are chosen on the individual chromosomes (strings) and the genetic material is exchanged at these points.

Option C) The mask in uniform crossover is a binary array of the same length as the chromosome. Each bit in the mask determines whether the corresponding bit in the offspring chromosome is inherited from the first parent (if the mask bit is 0) or the second parent (if the mask bit is 1).

For example, if the mask is [0, 1, 0, 1, 0, ...], then the offspring chromosome will inherit bits alternately from the first and second parents based on the mask.

Option D) In N-point crossover, each crossover point is set on the boundary between two prototypes in the chromosome.

So option C is the correct answer.

#### Q77 Text Solution:

A. The term "complement" typically refers to the representation of negative numbers in computer systems using two's complement or one's complement notation. This involves representing negative numbers by complementing the bits of the positive counterpart.

B. E- OR gates play a crucial role within adder circuits for specific logical operations necessary for binary addition and subtraction.

C. In a computer's ALU, the accumulator is a buffer that stores intermediary arithmetic logic unit outcomes.

D. The CPU's control unit decodes the program instruction once it is loaded into the instruction register (IR).

So option C is the correct answer.

#### Q78 Text Solution:

Option A:  $+-^* 2 / 8432$

$+-^* 2 2 3 2 (8/4 = 2)$

$+- 4 3 2 (2*2 = 4)$

$+ 1 2 (4-3 = 1)$

$3 (1+2 = 3)$

Option B:  $\uparrow -^* 33 * 425$



$\uparrow - * 3 \ 3 \ 8 \ 5 \ (4 * 2 = 8)$

$\uparrow - 9 \ 8 \ 5 \ (3 * 3 = 9)$

$\uparrow 1 \ 5 \ (9 - 8 = 1)$

$1 \ (1^5 = 1)$

Option C:  $+ - \uparrow 32 \uparrow 23 / 6 - 42$

$+ - 9 \ 8 / 6 \ 2 \ (3^2 = 9, 2^3 = 8, 4 - 2 = 2)$

$+ - 9 \ 8 \ 3 \ (6 / 2 = 3)$

$+ 1 \ 3 \ (9 - 8 = 1)$

$4 \ (1 + 3 = 4)$

Option D:  $* + 3 + 3 \uparrow 3 + 113$

$* + 3 + 3 \uparrow 3 \ 2 \ 3 \ (1 + 1 = 2)$

$* + 3 + 27 \ 2 \ 3 \ (3^3 = 27)$

$* + 3 \ 29 \ 3 \ (27 + 2 = 29)$

$* \ 32 \ 3 \ (29 + 3 = 32)$

$96 \ (32 * 3)$

So  $A = 3, B = 1, C = 4, D = 96$

therefore correct sequence will be B A C D,  
Option B.

#### Q79 Text Solution:

Option A) Degree distribution follows scale-free: Social networks follow this property. A scale-free network has a degree distribution that follows a power-law, meaning there are a few nodes (hubs) with a very high number of connections (high degree) and many nodes with fewer connections.

Option B) Average shortest path length is a global measure of how connected nodes are on average. Networks with shorter average paths are generally more connected. So option B is false.

Option C) In a directed social network, connections between nodes are directional. For example, in a Twitter network, Smith might follow Jones but that does not mean that Jones follows Smith. Each directional relationship would have an edge to represent it, typically with an arrow.

Option D) A social network is called a small-world network if any two of people in the network can reach each other through a short sequence of acquaintances.

So option B is the correct answer.

#### Q80 Text Solution:

For line drawing, the commonly used algorithms are:

A. DDA algorithm (Digital Differential Analyzer):

DDA algorithm is a basic algorithm for rasterization of lines. It calculates the intermediate points between two endpoints of a line based on the difference between the coordinates of these points.

B. Bresenham algorithm:

Bresenham's line algorithm is another rasterization algorithm that is widely used for drawing lines. It is efficient and calculates the positions of points on a line using integer arithmetic, which makes it suitable for implementations without floating-point unit.

C. Midpoint algorithm:

The midpoint algorithm is typically used for drawing circles and ellipses by determining the points that approximate the circle or ellipse shape.

D. Curve algorithm:

Curve algorithms, such as Bezier curves or B-spline curves, are used for drawing curves rather than straight lines. They involve more complex calculations compared to line-drawing algorithms.

Therefore, the algorithms used specifically for line drawing are:

A. DDA algorithm

B. Bresenham algorithm

Thus, the correct option is C.

#### Q81 Text Solution:

Option A: The Nyquist theorem states that to accurately reconstruct the original analog signal from its samples, the sampling rate  $f_s$  must be at least twice the highest frequency  $f_{\max}$  present in the signal.

Mathematically, this is expressed as  $f_s \geq 2 \times f_{\max}$

Here,  $f_s$  is the sampling rate (number of samples per second), and  $f_{\max}$  is the highest frequency component present in the analog signal.

Option (B) states that the sampling rate must be at least 2 times the lowest frequency contained in the signal. This is incorrect because the



Nyquist theorem pertains to the highest frequency, not the lowest.

Options (3) and (4) suggest sampling rates that are less than the Nyquist rate (less than 2 times the highest frequency), which would lead to inaccurate reconstruction of the analog signal.

So option A is the correct answer.

**Q82 Text Solution:**

$$A = \{(x_1, 0.3), (x_2, 0.7), (x_3, 0.4)\}$$

$$B = \{(x_1, 0.5), (x_2, 0.2), (x_3, 0.5)\}$$

$$A \cap B = \{(x_1, 0.3), (x_2, 0.2), (x_3, 0.4)\}$$

$$\mu_A \cap B(x_1) = \min(\mu_A(x_1), \mu_B(x_1))$$

$$= \min(0.3, 0.5) = 0.3$$

$$\mu_A \cap B(x_2) = 0.2 \text{ and } \mu_A \cap B(x_3) = 0.4$$

So option D is the correct answer.

**Q83 Text Solution:**

Unsigned 8-bit variable:

This type can represent values from 0 to  $2^8 - 1 = 255$ .

Therefore, the least value is 0.

Signed 8-bit variable:

For signed variables using 8 bits, the range is typically from  $-2^7$  to  $2^7 - 1$ .

This corresponds to -128 to 127

Therefore, the least value for a signed 8-bit variable is -128.

So option D is the correct answer.

**Q84 Text Solution:**

Explanation:

The following statements are True for a complete binary Tree:

A. It is always an odd number of vertices. This is True. For  $n > 1$ , a full binary tree must have an odd number of nodes (since each node has either 0 or 2 children, and we start with 1 root node). So,  $n$  should be odd.

B. With  $i$  internal vertices, it has  $i+1$  leaves. True. In a complete binary tree, the number of internal vertices  $i$  is equal to the number of leaves  $l$  plus one:  $l = i + 1$

This relationship holds because each internal vertex has exactly two children (except possibly the last level of the tree), and each leaf

corresponds to a node that does not have any children.

C. With  $l$  leaves, it has  $2l-1$  vertices.

**True. The total number of vertices  $v$  in a complete binary tree with  $l$  leaves is given**

**by:  $v = 2l - 1$**

D. With  $2n-1$  vertices, it has  $n$  leaves. True. For a complete binary tree with  $n$  leaves, the total number of vertices  $v$  is  $v = 2n - 1$ . This relationship holds because every complete binary tree with  $n$  leaves has  $2n - 1$  vertices in total.

So option D is the correct answer.

**Q85 Text Solution:**

Size of the memory needed:

Memory size required:  $2048 \times 8$  bytes.

Size of each chip:  $128 \times 4$  bytes.

Calculating the number of chips:

First, calculate the total number of bytes required for the memory:

Total bytes required =  $2048 \times 8 = 16384$  bytes

Now, calculate how many chips are needed to cover this total memory size:

Each chip provides  $128 \times 4$  bytes.

Calculate how many such chips are needed:

Number of chips = Total bytes required / Bytes per chip

$$= 16384 / 128 \times 4$$

$$\text{Number of chips} = 16384 / 512 = 32$$

So option C is the correct answer.

**Q86 Text Solution:**

Sequential circuits are circuits where the output not only depends on the current inputs but also on the past history of inputs. They contain memory elements such as flip-flops to store information and exhibit sequential behavior.

A. Flip flop: Flip-flops are fundamental memory elements in digital circuits used to store binary data. They are essential in sequential circuits for storing state information.

B. Multiplexer: A multiplexer is a combinational circuit that selects one of multiple input signals and forwards it to a single output based on control signals. It does not inherently store any



state information and operates purely based on current inputs.

C. Counter: A counter is a sequential circuit that counts in a predefined sequence and maintains a state. It uses flip-flops to store the current count and has sequential behavior.

D. Decoder: A decoder is a combinational circuit that converts a binary code into a set of outputs where only one of the outputs is active at a time based on the input code. It does not have memory elements and operates purely based on current inputs.

E. Adder: An adder is a combinational circuit that performs addition of binary numbers. It does not store any state information and operates purely based on current inputs.

So option B is the correct answer.

#### Q87 Text Solution:

White box testing refers to a testing technique where the internal structure, design, and implementation of the software under test are known to the tester. Let's analyze each option to determine which ones are considered white box testing techniques:

##### A. Path Testing:

Path testing is a white box testing technique where test cases are designed to execute all possible paths in a software application's source code.

It requires knowledge of the internal structure and logic of the code to ensure that every path is tested.

##### B. Loop Testing:

Loop testing is a white box testing technique that focuses on testing the functionality of loops within the code.

Test cases are designed based on the structure of loops and their boundaries, requiring knowledge of the loop structure and logic.

##### C. Regression Testing:

Regression testing is not specifically a white box testing technique. It is a type of testing that ensures previously developed and tested software still performs correctly after a change

or addition.

It does not require knowledge of the internal structure of the software but focuses on the overall functionality.

##### D. Non-functional Testing:

Non-functional testing is a type of testing that verifies the non-functional aspects of the software such as performance, reliability, usability, etc.

It is not specifically a white box testing technique as it does not require knowledge of the internal structure of the software.

So option A is the correct answer.

#### Q88 Text Solution:

##### A. Programmed I/O:

In programmed I/O, the CPU controls data transfer between external devices and memory directly using I/O instructions. This method involves the CPU actively waiting for each I/O operation to complete before proceeding, which can be inefficient for large data transfers.

##### B. Interrupt-driven I/O:

Interrupt-driven I/O allows the CPU to handle other tasks while waiting for I/O operations to complete. When an I/O operation finishes, the device sends an interrupt signal to the CPU, prompting it to handle the completed operation. This approach reduces CPU idle time and improves overall system efficiency.

##### D. Direct Memory Access (DMA):

DMA allows external devices to transfer data directly to or from memory without CPU intervention. A DMA controller manages the data transfer, freeing the CPU to perform other tasks. This method significantly reduces CPU overhead and speeds up data transfer rates between devices and memory.

##### C. Rotational I/O:

Rotational I/O is not a standard term for describing I/O techniques in general computing contexts. It refers to specific operations in disk drives where data is read or written as the disk rotates, but it's not commonly listed as a general



I/O technique.

E. Channelized I/O:

Channelized I/O involves using specialized hardware channels or controllers to manage I/O operations independently of the CPU. It's commonly used in mainframe and high-performance computing environments to offload I/O processing from the CPU, but it's less commonly categorized as a fundamental I/O technique compared to programmed I/O, interrupt-driven I/O, and DMA.

So option B is the correct answer.

#### Q89 Text Solution:

Option A) Execution of code: This refers to running the actual software code to test its functionality, performance, and other attributes.

This is a fundamental part of testing in SQA.

Option B) No execution of code: This refers to techniques such as static analysis, code reviews, inspections, and walkthroughs, where the focus is on examining the code or design without actually running the software. So this is included in the software quality assurance process. These techniques aim to identify issues and defects early in the development lifecycle before the software is executed.

Option C) Preventive technique: This refers to techniques and activities that aim to prevent defects from occurring in the software. This includes activities such as requirements analysis, design reviews, code reviews, and the use of coding standards and guidelines which include software quality assurance processes.

Option D) A corrective technique: This refers to techniques applied after defects or issues have been identified during testing or after deployment. Examples include debugging, rework, and patching to fix bugs that were discovered.

So option B is the correct answer.

#### Q90 Text Solution:

Convergence testing:

Explanation: Convergence testing in genetic algorithms involves determining when the

algorithm has reached an acceptable solution or when further optimization is unlikely to significantly improve the solution.

Computational Cost: Convergence testing generally involves iterating over the population and evaluating fitness metrics or checking conditions, which can be moderate in computational cost depending on the complexity of the fitness function and convergence criteria.

Initial population creation:

Explanation: In genetic algorithms, the initial population consists of randomly generated individuals (solutions) to start the optimization process.

Computational Cost: Generating the initial population involves creating a specified number of individuals, each with its own set of parameters or genes. While this operation can be computationally intensive, it is typically a one-time cost at the start of the algorithm.

Selection of sub-population for mating:

Explanation: During each generation of a genetic algorithm, a sub-population of individuals is selected for mating based on their fitness (usually through techniques like tournament selection, roulette wheel selection, etc.). This selection determines which individuals contribute to the next generation.

Computational Cost: Selecting individuals for mating involves evaluating fitness scores, sorting or selecting based on probabilities, and potentially maintaining diversity in the population. This operation is crucial but is generally less computationally expensive compared to other operations like crossover and mutation.

Reproduction to produce the next generation:

Explanation: Reproduction in genetic algorithms involves using selected individuals from the current population to create offspring (next generation) through crossover (combining genetic information from two parents) and mutation (introducing random changes).

Computational Cost: Reproduction typically involves iterating through selected individuals,



performing crossover and mutation operations, and creating new offspring. This operation can be computationally expensive, especially if the population size is large or if the genetic operations (crossover and mutation) are complex. So option D is the correct answer.

**Q91. Text Solution:**

Let  $x$  be the number of units of type A to be produced and  $y$  be the number of units of type B to be produced. It is given that:

|                     | Red Wool | Green Wool | Blue Wool | Profit |
|---------------------|----------|------------|-----------|--------|
| Type A              | 2 yard   | 3 yard     |           | Rs. 30 |
| Type B              | 3 yard   | 2 yard     | 2 yard    | Rs. 50 |
| (available at most) | 8 yard   | 10 yard    | 15 yard   |        |

Let  $Z$  be the profit function. The mathematical formulation of the given problem is  $\text{Max. } Z = 30x + 50y$  subject to the constraints:

$$2x + 3y \leq 8$$

$$3x + 2y \leq 10$$

$$0x + 2y \leq 15$$

Steps for formulation are summarised as

Step 1: write the given LPP in its standard form.

Step 2: identify the variables of dual problem which are same as the number of constraints equation.

Step 3: write the objective function of the dual problem by using the constants of the right hand side of the constraints.

Step 4: if primal is max/min type then dual is min/max type and the constraints are  $\geq/\leq$  type.

So, the objective function for its dual problem is:

$$\text{Min } P = 8u + 10v + 15w$$

Option B is the correct answer.

**Q92. Text Solution:**

Let  $x$  be the number of units of type A to be produced and  $y$  be the number of units of type B to be produced. It is given that:

|        | Red Wool | Green Wool | Blue Wool | Profit |
|--------|----------|------------|-----------|--------|
| Type A | 2 yard   | 3 yard     |           | Rs. 30 |

|                     |        |         |         |        |
|---------------------|--------|---------|---------|--------|
| Type B              | 3 yard | 2 yard  | 2 yard  | Rs. 50 |
| (available at most) | 8 yard | 10 yard | 15 yard |        |

Let  $Z$  be the profit function. The mathematical formulation of the given

problem is  $\text{Max. } Z = 30x + 50y$

Thus Option C is the correct answer.

**Q93. Text Solution:**

The mathematical formulation of the given problem is  $\text{Max. } Z = 30x + 50y$  subject to the constraints:

$$2x + 3y = 8 \text{ ---(i)}$$

$$3x + 2y = 10 \text{ ---(ii)}$$

$$0x + 2y = 15 \text{ ---(iii)}$$

Multiply equation (i) by 2 and (ii) by 3

$$4x + 6y = 16 \text{ ---(iv)}$$

$$9x + 6y = 30 \text{ ---(v)}$$

Subtract eq (iv) from (v)

$$5x = 14$$

$$x = 2.8$$

So unit length used for cloth A when income is maximized. Thus option C is the correct answer.

**Q94. Text Solution:**

Let  $x$  be the number of units of type A to be produced and  $y$  be the number of units of type B to be produced. It is given that:

|                     | Red Wool | Green Wool | Blue Wool | Profit |
|---------------------|----------|------------|-----------|--------|
| Type A              | 2 yard   | 3 yard     |           | Rs. 30 |
| Type B              | 3 yard   | 2 yard     | 2 yard    | Rs. 50 |
| (available at most) | 8 yard   | 10 yard    | 15 yard   |        |

subject to the constraints:

$$2x + 3y \leq 8$$

$$3x + 2y \leq 10$$

$$0x + 2y \leq 15$$

So the option B is the correct answer.

**Q95. Text Solution:**

The mathematical formulation of the given problem is  $\text{Max. } Z = 30x + 50y$

subject to the constraints:

$$2x + 3y = 8 \text{ ---(i)}$$



$$3x+2y = 10 \text{ ---(ii)}$$

$$0x+2y = 15 \text{ ---(iii)}$$

Multiply equation (i) by 2 and (ii) by 3

$$4x+6y = 16 \text{ ---(iv)}$$

$$9x+6y = 30 \text{ ---(v)}$$

Subtract eq (iv) from (v)

$$5x = 14$$

$$x = 2.8$$

Putting the value of x in eq (i)  $y = 0.8$

By putting the values of x and y in Z we get

$$30 \times 2.8 + 50 \times 0.8 = 84 + 40 = 124$$

So option B is the correct answer.

**Q96. Text Solution:**

Query (1): Select COUNT(\*) As total\_orders from orders where customer\_id = [Customer\_ID];

Explanation: This query counts all rows where the customer\_id matches [Customer\_ID], effectively counting all orders placed by the specific customer.

Query (2): Select COUNT(order\_id) As total\_orders from orders where customer\_id = [Customer\_ID];

Explanation: This query counts the non-null order\_id values where the customer\_id matches [Customer\_ID] for retrieving the total number of orders placed by a specific customer. It counts specifically the order\_id values, which should also accurately reflect the number of orders for the customer if order\_id is not null.

Query (3): Select COUNT(DISTINCT order\_id) As total\_orders from orders Where customer\_id = [Customer\_ID];

Explanation: This query counts the distinct order\_id values where the customer\_id matches [Customer\_ID].

While this query counts distinct order\_id values, it may not accurately reflect the total number of orders placed by the customer if there are duplicate order\_id values for the same customer. Therefore, it may not always return the correct total number of orders.

Query (4): Select SUM(order\_id) As total\_orders from orders where customer\_id = [Customer\_ID];

Explanation: This query attempts to sum up all

order\_id values where the customer\_id matches [Customer\_ID].

So option B is the correct answer.

**Q97. Text Solution:**

Option A) Each customer can place only one order.

Explanation: This statement implies that each customer is limited to placing only one order. This is false. Customers typically can place multiple orders over time.

Option B) Each sales representative can manage multiple customer accounts.

Explanation: This statement suggests that sales representatives can handle more than one customer account. This is true. Because each customer can place multiple orders and each order can contain multiple products. Each order is associated with a specific sales representative who manages the accountable products.

Option C) Each order can be associated with only one product.

Explanation: This statement implies that an order can include only one product. This is false. Orders include multiple products. A typical database design for orders would include a relationship where an order can have multiple line items, each representing a product purchased.

Statement (4): The database does not include a table for products.

Explanation: This statement suggests that there is no table in the database specifically for storing products. Without a table for products, it would be impossible to accurately track and manage products in the system.

So option B is the correct answer.

**Q98. Text Solution:**

Orders Table

In the database system described, where each customer can place multiple orders and each order can contain multiple products, the Orders table is indeed likely to have foreign keys referencing other tables in the database.

Specifically:



Customer\_ID (Foreign Key): The Orders table would typically include a foreign key referencing the Customers table. This foreign key establishes the relationship between orders and customers, indicating which customer placed each order.

Product\_ID (Foreign Key): If the design includes an intermediate table (like Order\_Details), the Orders table may have foreign keys referencing the Products table. This allows each order to list multiple products associated with it.

Sales\_Rep\_ID (Foreign Key): As mentioned, each order is associated with a specific sales representative. Therefore, the Orders table may also include a foreign key referencing the Sales Representatives table, indicating which sales representative is accountable for each order. So option B is the correct answer.

#### Q99. Text Solution:

Query (1): Select COUNT(DISTINCT customer\_id) As total\_customers From orders;

Explanation: This query counts the distinct customer\_id values from the orders table, which would give us the number of unique customers who have placed orders.

Query (2): Select COUNT(\*) As total\_customers from customers;

Explanation: This query counts all rows in the customer's table, which would give us the total number of customers in the database.

Query (3): Select COUNT(DISTINCT order\_id) As total\_customers from orders;

Explanation: This query counts the distinct order\_id values from the orders table. It does not relate to counting distinct customers who have placed orders.

Query (4): Select COUNT(DISTINCT customer\_id) As total\_customers from customers;

Explanation: This query counts the distinct customer\_id values from the customer's table. It gives us the total number of unique customers in the customer's table.

So option A is the correct answer.

#### Q100. Text Solution:

- Each customer can place multiple orders: This implies that for every customer in the "Customers" table, there can be multiple corresponding rows (orders) in the "Orders" table.
- Each order is associated with a specific customer: This indicates that there is a direct link from each row (order) in the "Orders" table back to a single customer in the "Customers" table.
- Many-to-One accurately describes the relationship where many orders (from the "orders" table) can be associated with exactly one customer (from the "Customers" table).

This relationship aligns with the scenario described, where each customer can place multiple orders over time, but each order is linked back to a single customer.

So option C is the correct answer.

